



UNIVERSITÀ
DI TORINO



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Fock-Exchange in Solids with a Local Basis Set: Hybrid DFAs Made Efficient

Giacomo Ambrogio

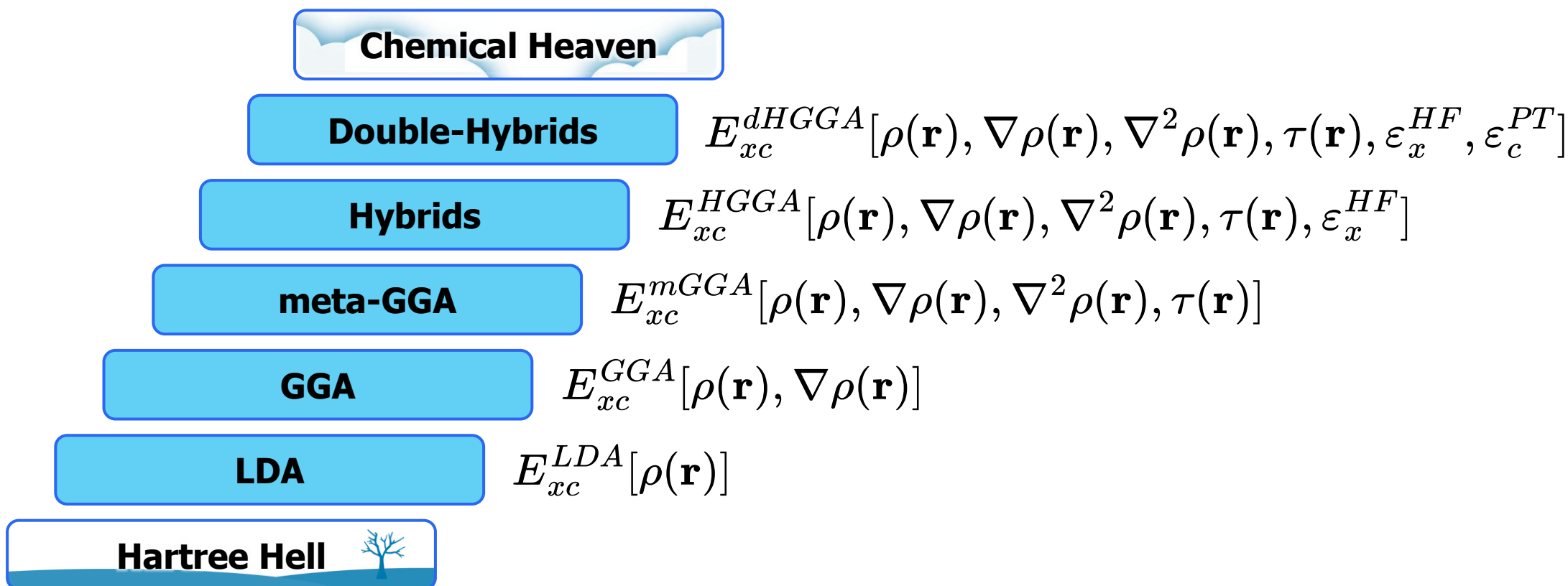
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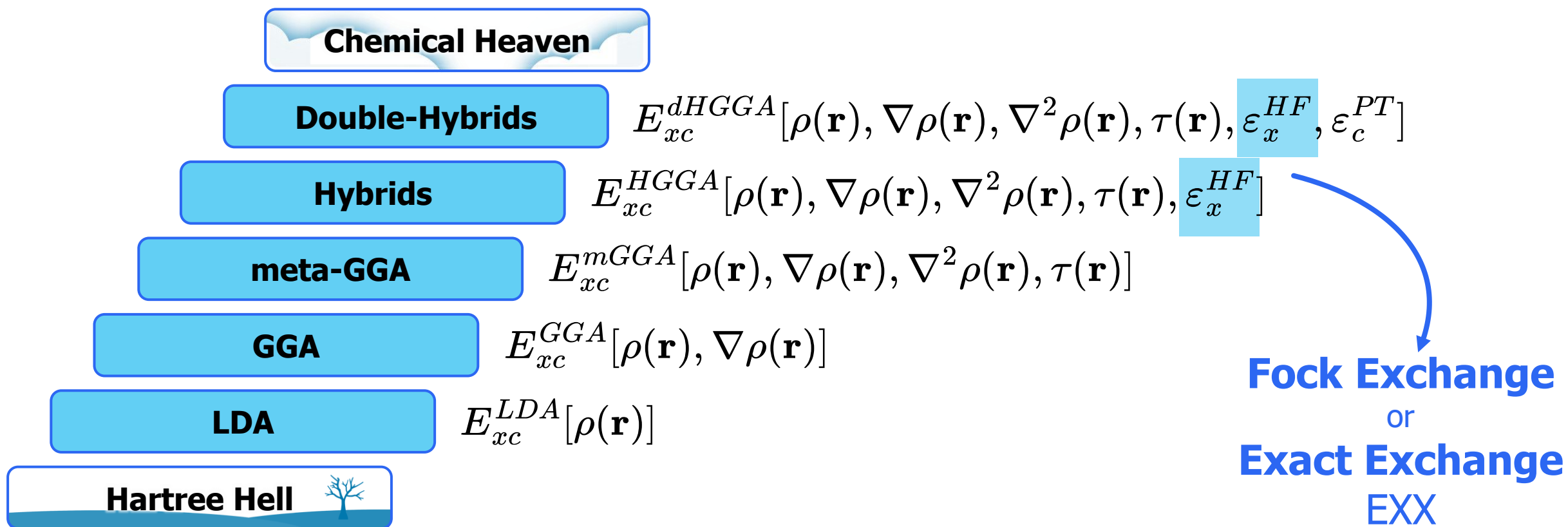
Molecular Biotechnology Center

Ab Initio Modelling in Solid State Chemistry

Jacobs Ladder



Jacobs Ladder



Replace a Fraction α of
DFT Exchange
with **Exact Exchange**

$$E_{xc} = E_c + E_x$$
$$E_{xc} = E_c + (1 - \alpha)E_x + \alpha E_{EXX}$$

Fixed α Parameter

$$(\chi_\mu \chi_\nu | \chi_\sigma \chi_\rho) = \int \int d\mathbf{r}_1 d\mathbf{r}_2 \frac{\chi_\mu(\mathbf{r}_1) \chi_\nu(\mathbf{r}_1) \chi_\sigma(\mathbf{r}_2) \chi_\rho(\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|}$$

Functional	α
B3LYP	20%
PBE0	25%
B1WC	16%
mPW1K	42.8%

Range Separated Hybrids

$$(\chi_\mu\chi_\nu|\chi_\sigma\chi_\rho) = \int \int d\mathbf{r}_1 d\mathbf{r}_2 \frac{\chi_\mu(\mathbf{r}_1)\chi_\nu(\mathbf{r}_1)\chi_\sigma(\mathbf{r}_2)\chi_\rho(\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|}$$

Range Separated Hybrids

$$(\chi_\mu \chi_\nu | \chi_\sigma \chi_\rho) = \iint d\mathbf{r}_1 d\mathbf{r}_2 \frac{\chi_\mu(\mathbf{r}_1) \chi_\nu(\mathbf{r}_1) \chi_\sigma(\mathbf{r}_2) \chi_\rho(\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|}$$

$$|\mathbf{r}_1 - \mathbf{r}_2| = r$$

$$\frac{1}{r} = \underbrace{\frac{\text{erfc}(\omega r)}{r}}_{\text{Short Range}} + \underbrace{\frac{\text{erf}(\omega r)}{r}}_{\text{Long Range}}$$

Error Function

ω
Adjustable Parameter

Range Separated Hybrids

$$(\chi_\mu \chi_\nu | \chi_\sigma \chi_\rho) = \iint d\mathbf{r}_1 d\mathbf{r}_2 \frac{\chi_\mu(\mathbf{r}_1) \chi_\nu(\mathbf{r}_1) \chi_\sigma(\mathbf{r}_2) \chi_\rho(\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|}$$

$|\mathbf{r}_1 - \mathbf{r}_2| = r$

$$\frac{1}{r} = \underbrace{\frac{\text{erfc}(\omega r)}{r}}_{\text{Short Range}} + \underbrace{\frac{\text{erf}(\omega r)}{r}}_{\text{Long Range}}$$

Error Function

ω
Adjustable Parameter

$$E_x = (1 - \alpha^{SR})E_x^{SR} + \alpha^{SR}E_{EXX}^{SR} + (1 - \alpha^{LR})E_x^{LR} + \alpha^{LR}E_{EXX}^{LR}$$

Functional	α^{SR}	α^{LR}
HSE06	25%	0%
wB97X	16.7%	100%
CAM-B3LYP	19%	65%

Range Separated Hybrids

$$(\chi_\mu \chi_\nu | \chi_\sigma \chi_\rho) = \iint d\mathbf{r}_1 d\mathbf{r}_2 \frac{\chi_\mu(\mathbf{r}_1) \chi_\nu(\mathbf{r}_1) \chi_\sigma(\mathbf{r}_2) \chi_\rho(\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|}$$

$|\mathbf{r}_1 - \mathbf{r}_2| = r$

$$\frac{1}{r} = \underbrace{\frac{\text{erfc}(\omega r)}{r}}_{\text{Short Range}} + \underbrace{\frac{\text{erf}(\omega r)}{r}}_{\text{Long Range}}$$

Error Function

ω
Adjustable Parameter

$$E_x = (1 - \alpha^{SR})E_x^{SR} + \alpha^{SR}E_{EXX}^{SR} + (1 - \alpha^{LR})E_x^{LR} + \alpha^{LR}E_{EXX}^{LR}$$

$E_x = \text{Short Range} + \text{Mid Range} + \text{Long Range}$

Functional	α^{SR}	α^{LR}
HSE06	25%	0%
wB97X	16.7%	100%
CAM-B3LYP	19%	65%

Alpha in Hybrids

$$E_{xc} = E_c + (1 - \alpha)E_x + \alpha E_{EXX}$$

In CRYSTAL you can set
the α Parameter in Input

```
DFT
PBE
HYBRID
15
END
```

Tuned α Parameter

Alpha in Hybrids

$$E_{xc} = E_c + (1 - \alpha)E_x + \alpha E_{EXX}$$

In CRYSTAL you can set the α Parameter in Input

```
DFT
PBE
HYBRID
15
END
```

Tuned α Parameter

Functional

DD-PBE0

α

$1/\epsilon_\infty$

α Dependent from the Dielectric Constant of the Material

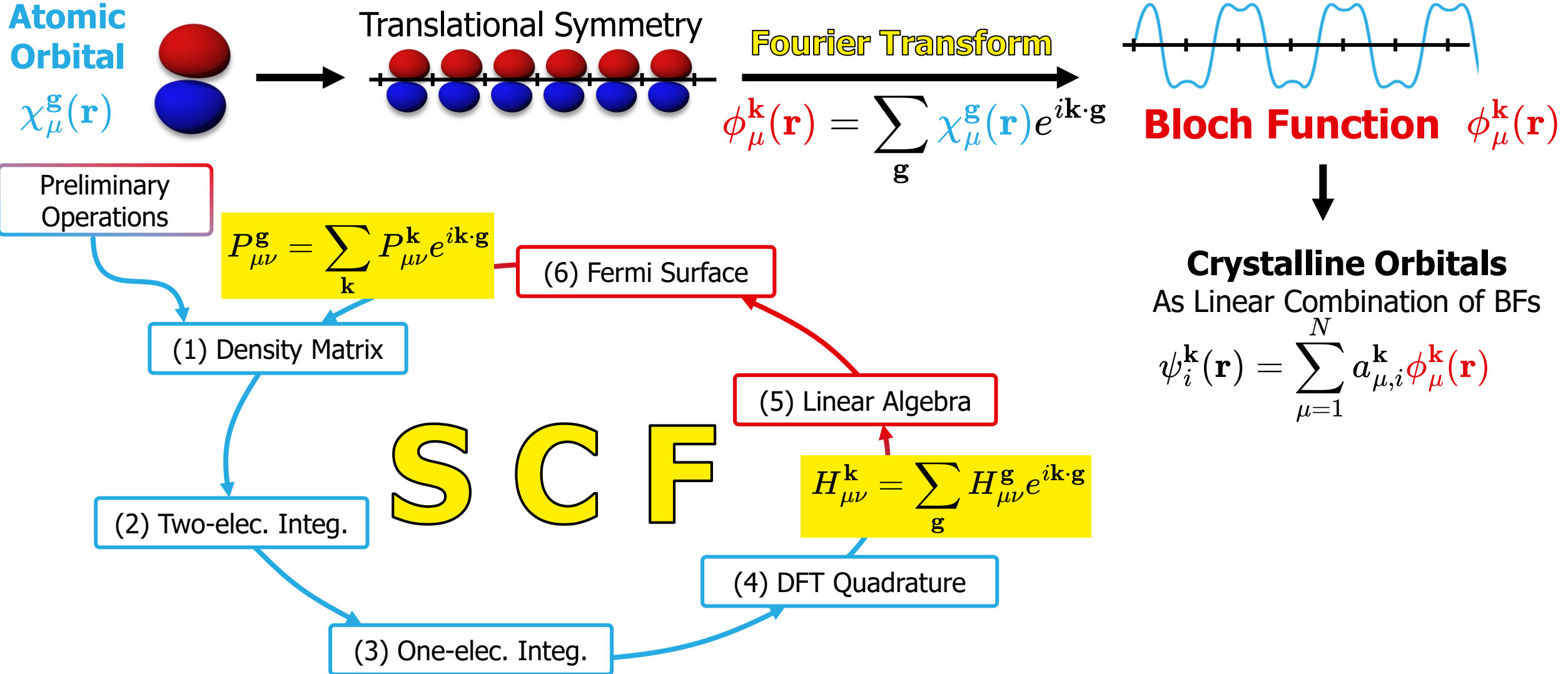
$$\alpha = \frac{1}{\epsilon_\infty}$$

- From Experiments
- From First Principles

Fully Automated Procedure in CRYSTAL

```
... geometry section ...
SCHYBRID
END
ENDgeom
```

Local Basis Set Approach



Exchange and Correlation Matrix

$$\mathbf{H}^g = \mathbf{T}^g + \mathbf{Z}^g + \mathbf{C}^g + \mathbf{V}_{\text{xc}}^g$$

$$[\mathbf{V}_{\text{xc}}^g]_{\mu\nu} = \frac{\partial E_{\text{xc}}[\{\psi_i\}]}{\partial P_{\mu\nu}^g}$$

Generalized Kohn-Sham
Framework

Exact Exchange

Global Hybrid DFA: $E_{\text{xc}}[\{\psi_i\}] = \int d\mathbf{r} \varepsilon_c + (1 - \alpha) \int d\mathbf{r} \varepsilon_x + \alpha \underline{E_{\text{exx}}[\{\psi_i\}]}$

$$E_{\text{exx}} = -\frac{1}{2} \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} | \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

$$(\chi_{\mu}^0 \chi_{\nu}^{\mathbf{s}} | \chi_{\sigma}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}}) = \iint d\mathbf{r}_1 d\mathbf{r}_2 \frac{\chi_{\mu}^0(\mathbf{r}_1) \chi_{\nu}^{\mathbf{s}}(\mathbf{r}_1) \chi_{\sigma}^{\mathbf{g}}(\mathbf{r}_2) \chi_{\rho}^{\mathbf{s}+\mathbf{q}}(\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|}$$

Computing exx in CRYSTAL

$$E_{\text{exx}} = -\frac{1}{2} \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} | \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

Summations over
7 Indices

Computing e_{xx} in CRYSTAL

$$E_{\text{exx}} = -\frac{1}{2} \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} | \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

Summations over
7 Indices

- Finite summation over all the orbitals (in the reference cell)

Computing e_{xx} in CRYSTAL

$$E_{\text{exx}} = -\frac{1}{2} \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} | \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

Summations over
7 Indices

- Finite summation over all the orbitals (in the reference cell)
- Infinite summations over direct space vectors

Truncation
Based on *Orbital Overlap*

Computing e_{xx} in CRYSTAL

$$E_{\text{exx}} = -\frac{1}{2} \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} | \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

Summations over
7 Indices

- Finite summation over all the orbitals (in the unit cell)
- Infinite summations over direct space vectors

Truncation
Based on *Orbital Overlap*

• **s** Summation

$$\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} \quad \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}}$$

If either of the two orbitals couple
($\mu\sigma$ or $\nu\rho$) have *low overlap*,
the product is close to zero

We Discard the Integral

Computing exx in CRYSTAL

$$E_{\text{exx}} = -\frac{1}{2} \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} | \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

Summations over
7 Indices

- Finite summation over all the orbitals (in the unit cell)
- Infinite summations over direct space vectors

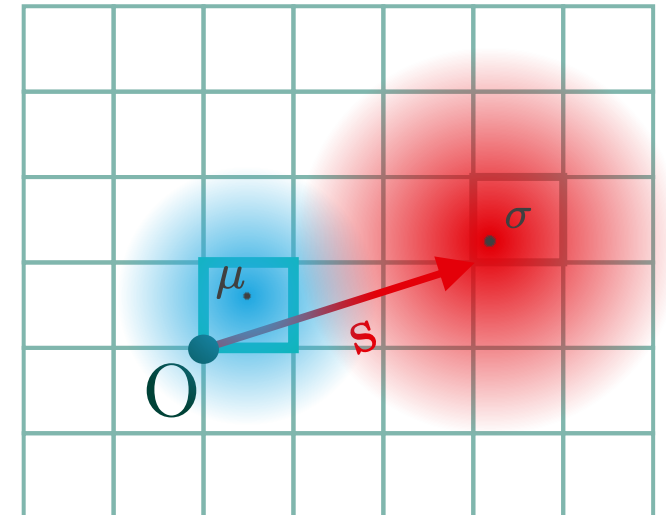
Truncation
Based on *Orbital Overlap*

• **s** Summation

$$\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} \quad \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}}$$

S_s **Threshold** for
s Summation

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7 7 **7** 15 30



Computing exx in CRYSTAL

$$E_{\text{exx}} = -\frac{1}{2} \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} | \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

Summations over
7 Indices

- Finite summation over all the orbitals (in the unit cell)
- Infinite summations over direct space vectors

Truncation
Based on *Orbital Overlap*

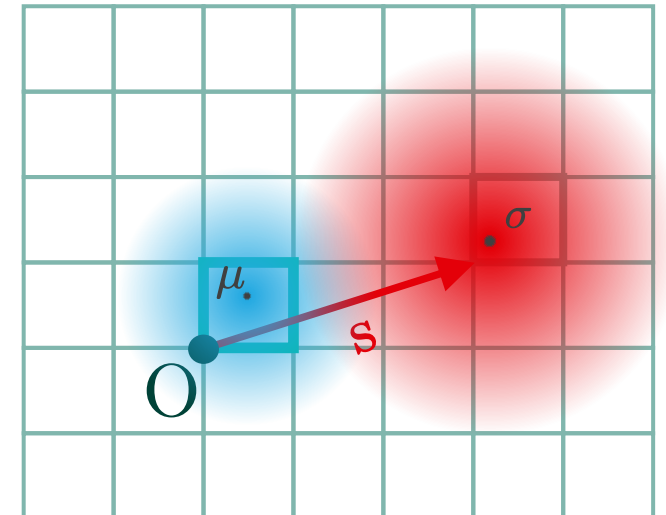
• **s** Summation

S_s **Threshold** for
s Summation

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7 7 **7** 15 30

$$\underbrace{\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}}}_{\text{red arrow}} \quad \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}}$$

$$\chi_{\mu}^0 \chi_{\nu}^{\mathbf{s}} > S_s \text{ KEEP}$$



Computing exx in CRYSTAL

$$E_{\text{exx}} = -\frac{1}{2} \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} | \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

Summations over
7 Indices

- Finite summation over all the orbitals (in the unit cell)
- Infinite summations over direct space vectors

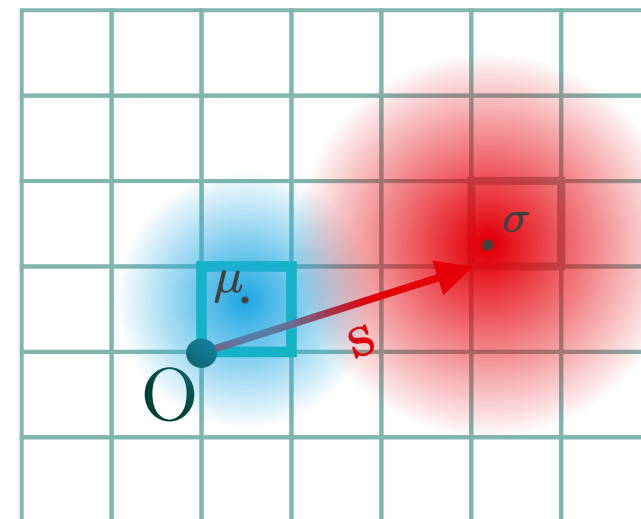
Truncation
Based on *Orbital Overlap*

• **s** Summation

S_s **Threshold** for
s Summation

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7 7 **7** 15 30

$$\begin{aligned} & \chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} & \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}} \\ & \downarrow \\ & \chi_{\mu}^0 \chi_{\nu}^{\mathbf{s}} > S_s & \text{KEEP} \\ & \chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} < S_s & \text{DISCARD} \end{aligned}$$



Computing e_{xx} in CRYSTAL

$$E_{\text{exx}} = -\frac{1}{2} \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} | \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

Summations over
7 Indices

- Finite summation over all the orbitals (in the unit cell)
- Infinite summations over direct space vectors

Truncation
Based on *Orbital Overlap*

• **s** Summation

S_s **Threshold** for
s Summation

$$\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} \quad \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}}$$

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Same evaluation for
the $\nu\rho$ couple

If either is below S_s , this **s**
(and all greater values)
is discarded.

Computing e_{xx} in CRYSTAL

$$E_{\text{exx}} = -\frac{1}{2} \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} | \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

Summations over
7 Indices

- Finite summation over all the orbitals (in the unit cell)
- Infinite summations over direct space vectors

Truncation
Based on *Orbital Overlap*

- **s** Summation
- **g** and **q** Summations

$P_{\mu\nu}^{\mathbf{g}}$

$P_{\sigma\rho}^{\mathbf{q}}$

Truncated for **decay of density matrix elements**

Computing exx in CRYSTAL

$$E_{\text{exx}} = -\frac{1}{2} \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} | \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

Summations over **7 Indices**

- Finite summation over all the orbitals (in the unit cell)
- Infinite summations over direct space vectors

Truncation
Based on *Orbital Overlap*

- **s** Summation
- **g** and **q** Summations

$$\chi_{\mu}^0 \chi_{\nu}^{\mathbf{g}} < S_g \quad \chi_{\sigma}^0 \chi_{\rho}^{\mathbf{q}} < S_q$$

Truncated for **decay of density matrix elements**

S_g **Threshold** for **g** Summation

S_q **Threshold** for **q** Summation

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Use pseudo-overlap

- Like other criteria (**s** sum. and Coulomb)
- Take account for small/big shells

Computing exx in CRYSTAL

$$E_{\text{exx}} = -\frac{1}{2} \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} | \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

Summations over **7 Indices**

- Finite summation over all the orbitals (in the unit cell)
- Infinite summations over direct space vectors

Truncation
Based on *Orbital Overlap*

- **s** Summation
- **g** and **q** Summations

$$\chi_{\mu}^0 \chi_{\nu}^{\mathbf{g}} < S_g \quad \chi_{\sigma}^0 \chi_{\rho}^{\mathbf{q}} < S_q$$

Truncated for **decay of density matrix elements**

S_g **Threshold** for **g** Summation

S_q **Threshold** for **q** Summation

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7 7 7 **15 30**

Density matrix decay

- Dependent from the system
- Evolves during the SCF

Use pseudo-overlap

- Like other criteria (**s** sum. and Coulomb)
- Take account for small/big shells

Computing e_{xx} in CRYSTAL

$$E_{\text{exx}} = -\frac{1}{2} \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} | \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

Summations over **7 Indices**

- Finite summation over all the orbitals (in the unit cell)
- Infinite summations over direct space vectors

Truncation
Based on *Orbital Overlap*

- **s** Summation
- **g** and **q** Summations

Density matrix decay

- Dependent from the system
- Evolves during the SCF

Insulator

Overlap decays exponentially as **P**

Metal

Overlap decays exponentially, but not **P** (slower decay)

Use pseudo-overlap

- Like other criteria (**s** sum. and Coulomb)
- Take account for small/big shells

Computing e_{xx} in CRYSTAL

$$E_{\text{exx}} = -\frac{1}{2} \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} | \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

Summations over
7 Indices

- Finite summation over all the orbitals (in the unit cell)
- Infinite summations over direct space vectors

Truncation
Based on *Orbital Overlap*

- **s** Summation
- **g** and **q** Summations

Lower tolerances should be used to grant convergence

Density matrix decay

- Dependent from the system
- Evolves during the SCF

Insulator

Overlap decays exponentially as **P**

Metal

Overlap decays exponentially, but not **P** (slower decay)

Use pseudo-overlap

- Like other criteria (**s** sum. and Coulomb)
- Take account for small/big shells

Selection of Thresholds

$$\chi_{\mu}^0 \chi_{\nu}^{\mathbf{g}} < S_g$$

$$\chi_{\sigma}^0 \chi_{\rho}^{\mathbf{q}} < S_q$$

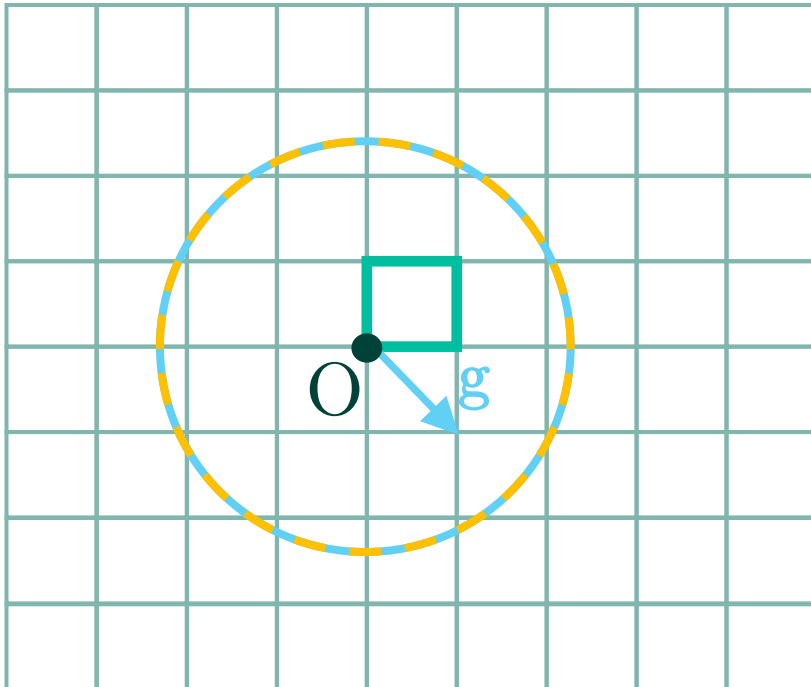
$$E_{\text{exx}} = -\frac{1}{2} \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} | \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

Consider Maximum Contribution in $\mathbf{g}$ Summation:

$$\sigma = \mu, \rho = \nu, \text{ and } \mathbf{s} = \mathbf{0} \quad (\chi_{\mu}^0 \chi_{\mu}^0 | \chi_{\nu}^{\mathbf{g}} \chi_{\nu}^{\mathbf{q}})$$

Non Zero Only for those $\mathbf{q}$ in the vicinity of $\mathbf{g}$

$\mathbf{g}$ and $\mathbf{q}$ Criteria are both centered on **Origin**



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$$S_g = S_q$$

Selection of Thresholds

$$\chi_{\mu}^0 \chi_{\nu}^{\mathbf{g}} < S_g$$

$$\chi_{\sigma}^0 \chi_{\rho}^{\mathbf{q}} < S_q$$

$$E_{\text{exx}} = -\frac{1}{2} \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} | \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

Consider Maximum Contribution in $\mathbf{g}$ Summation:

$$\sigma = \mu, \rho = \nu, \text{ and } \mathbf{s} = \mathbf{0} \quad (\chi_{\mu}^0 \chi_{\mu}^0 | \chi_{\nu}^{\mathbf{g}} \chi_{\nu}^{\mathbf{q}})$$

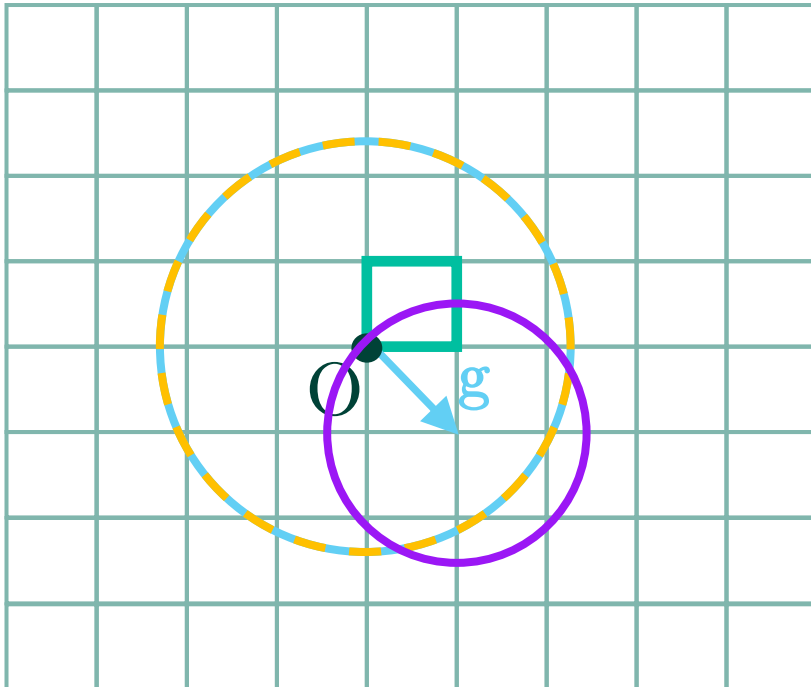
Non Zero Only for those $\mathbf{q}$ in the vicinity of $\mathbf{g}$

$\mathbf{g}$ and $\mathbf{q}$ Criteria are both centered on **Origin**

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$$S_g = S_q$$

Need to consider larger $\mathbf{q}$ threshold
to **Not Exclude** relevant
contribution for large $\mathbf{g}$ vectors



Selection of Thresholds

$$\chi_{\mu}^0 \chi_{\nu}^{\mathbf{g}} < S_g$$

$$\chi_{\sigma}^0 \chi_{\rho}^{\mathbf{q}} < S_q$$

$$E_{\text{exx}} = -\frac{1}{2} \sum_{\mu\nu} \sum_{\mathbf{g}} P_{\mu\nu}^{\mathbf{g}} \sum_{\sigma\rho} \sum_{\mathbf{q}} P_{\sigma\rho}^{\mathbf{q}} \sum_{\mathbf{s}} (\chi_{\mu}^0 \chi_{\sigma}^{\mathbf{s}} | \chi_{\nu}^{\mathbf{g}} \chi_{\rho}^{\mathbf{s}+\mathbf{q}})$$

Consider Maximum Contribution in $\mathbf{g}$ Summation:

$$\sigma = \mu, \rho = \nu, \text{ and } \mathbf{s} = \mathbf{0} \quad (\chi_{\mu}^0 \chi_{\mu}^0 | \chi_{\nu}^{\mathbf{g}} \chi_{\nu}^{\mathbf{q}})$$

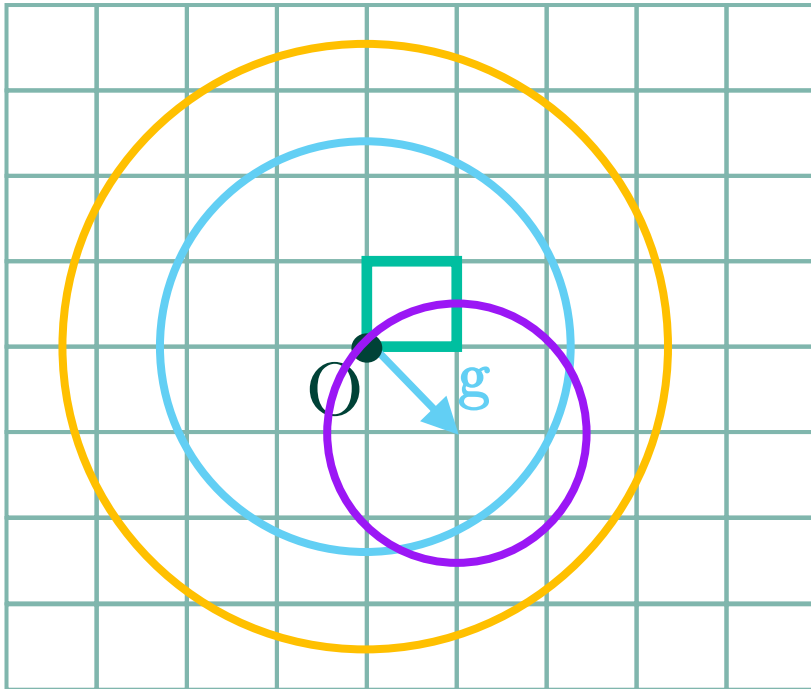
Non Zero Only for those $\mathbf{q}$ in the vicinity of $\mathbf{g}$

$\mathbf{g}$ and $\mathbf{q}$ Criteria are both centered on **Origin**

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7 7 7 **15 30**

$$S_g > S_q$$

Need to consider larger $\mathbf{q}$ threshold
to **Not Exclude** relevant
contribution for large $\mathbf{g}$ vectors



Bipolar Expansion

$$(\chi_\mu^0 \chi_\nu^s | \chi_\sigma^g \chi_\rho^{s+q}) = \iint d\mathbf{r}_1 d\mathbf{r}_2 \frac{\chi_\mu^0(\mathbf{r}_1) \chi_\nu^s(\mathbf{r}_1) \chi_\sigma^g(\mathbf{r}_2) \chi_\rho^{s+q}(\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|}$$


$$\frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|}$$

Bipolar Expansion in
terms of **Spherical**
Harmonics

$$\rho_1(\mathbf{s}_1; \mathbf{r}_1) = \chi_\mu(\mathbf{r}_1) \chi_\nu(\mathbf{r}_1)$$

$$\rho_2(\mathbf{s}_2; \mathbf{r}_2) = \chi_\sigma(\mathbf{r}_2) \chi_\rho(\mathbf{r}_2)$$

For Well-Separated
Non-Overlapping Distributions

Double Integral → Product of Single Integrals
(Reduced Computational Cost)

$$(\chi_\mu \chi_\nu | \chi_\sigma \chi_\rho) = \sum_{l_1, l_2} \sum_{m_1, m_2} B(l_1, m_1, l_2, m_2) \frac{Y_l^{-m}(\mathbf{s}_2; \mathbf{s}_1)}{R^{2l+1}} \left[\int d\mathbf{r}_1 \rho_1(\mathbf{s}_1; \mathbf{r}_1) Y_{l_1}^{m_1}(\mathbf{s}_1; \mathbf{r}_1) \right] \left[\int d\mathbf{r}_2 \rho_2(\mathbf{s}_2; \mathbf{r}_2) Y_{l_2}^{m_2}(\mathbf{s}_2; \mathbf{r}_2) \right]$$

Overlap Based
Criterion

S_b

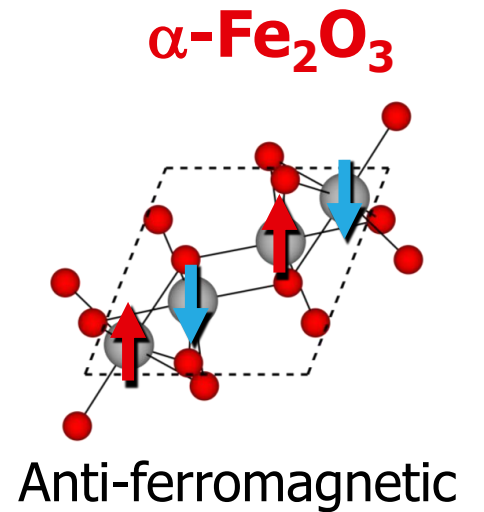
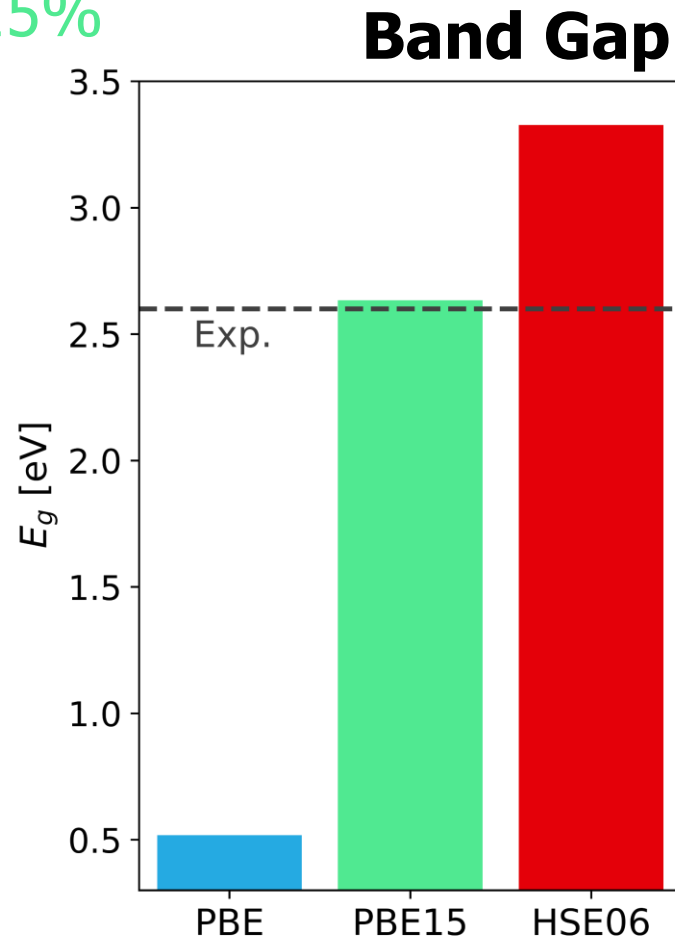
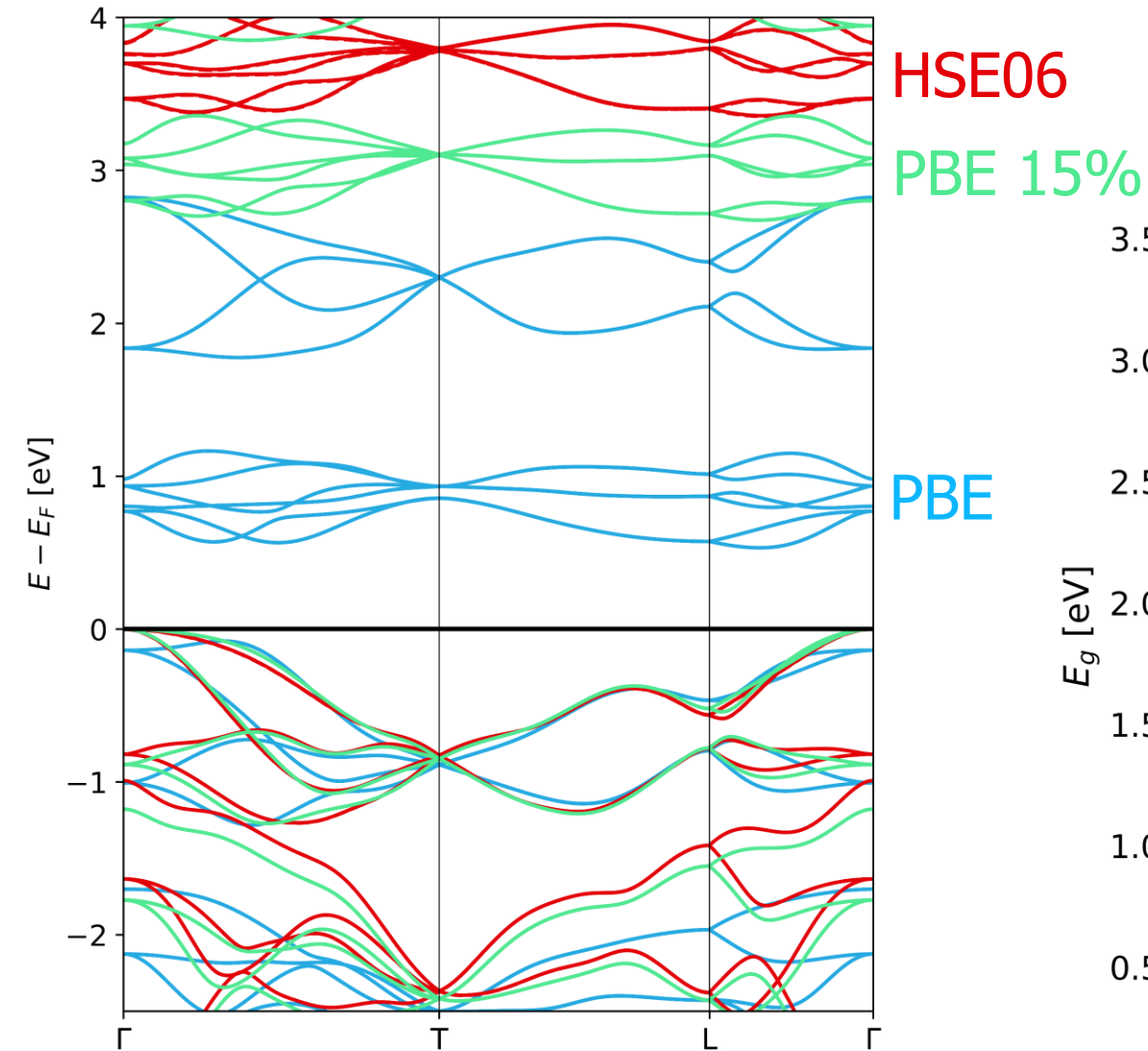
Integral is:

- Truncated
- Computed with Bipolar Approx.
- Computed Exactly

BIPOLAR

18 14

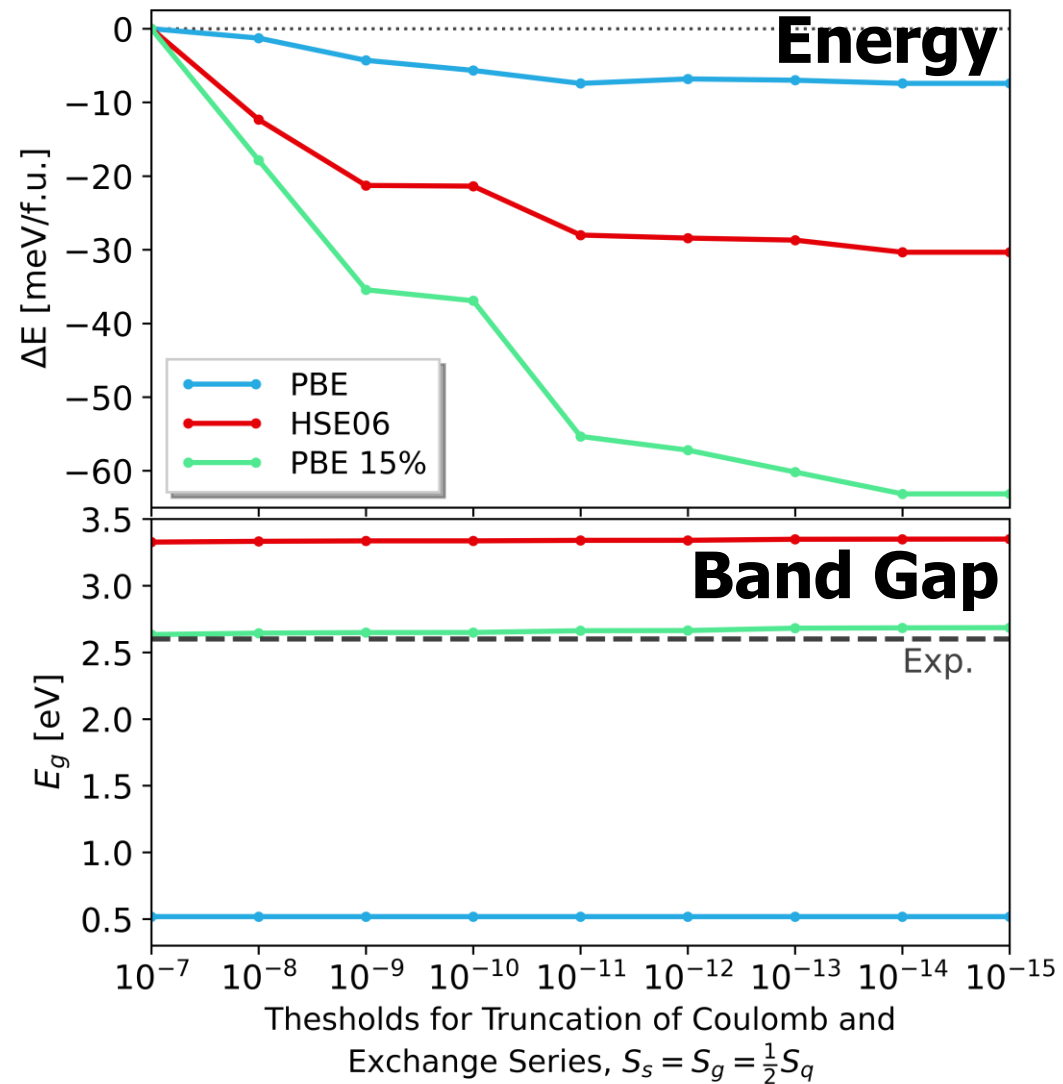
Hematite: Electronic Structure



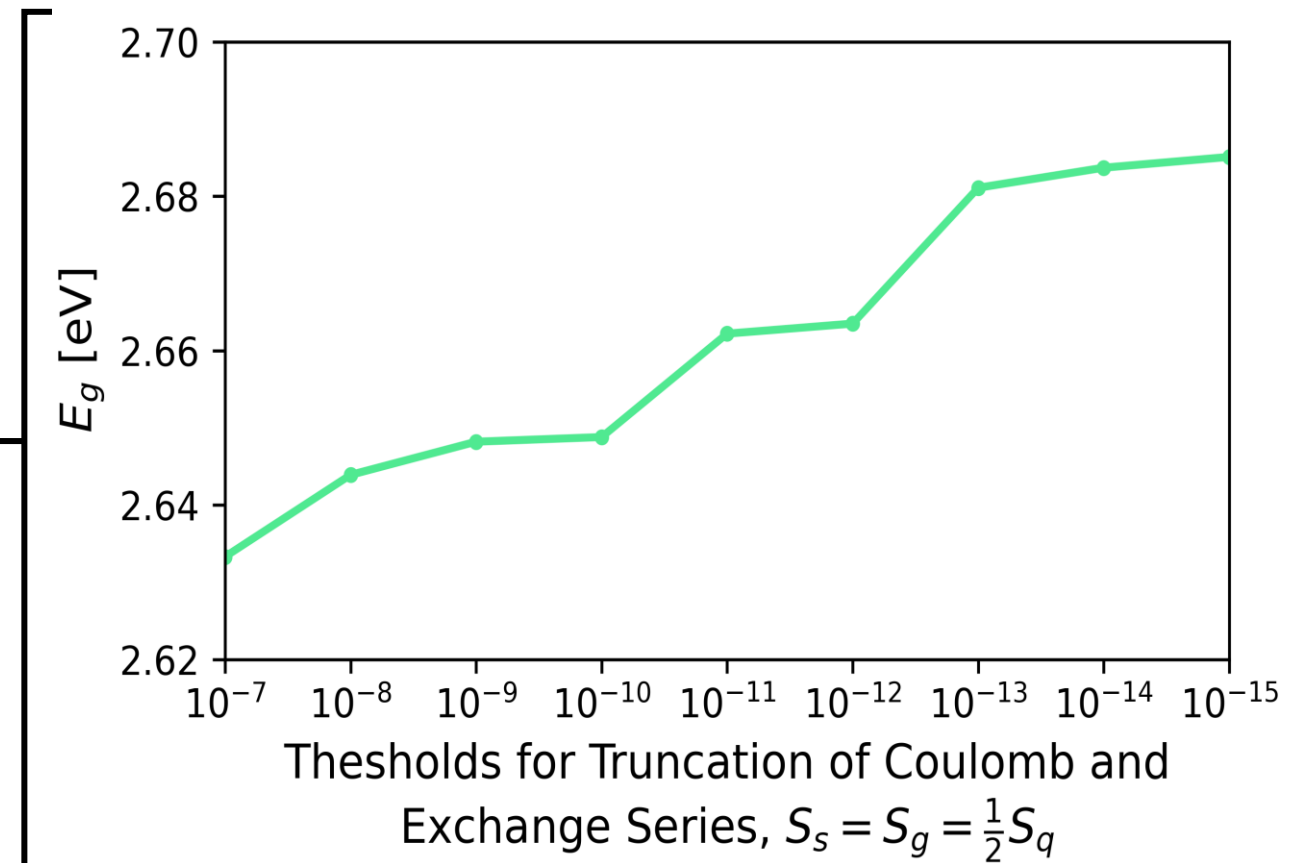
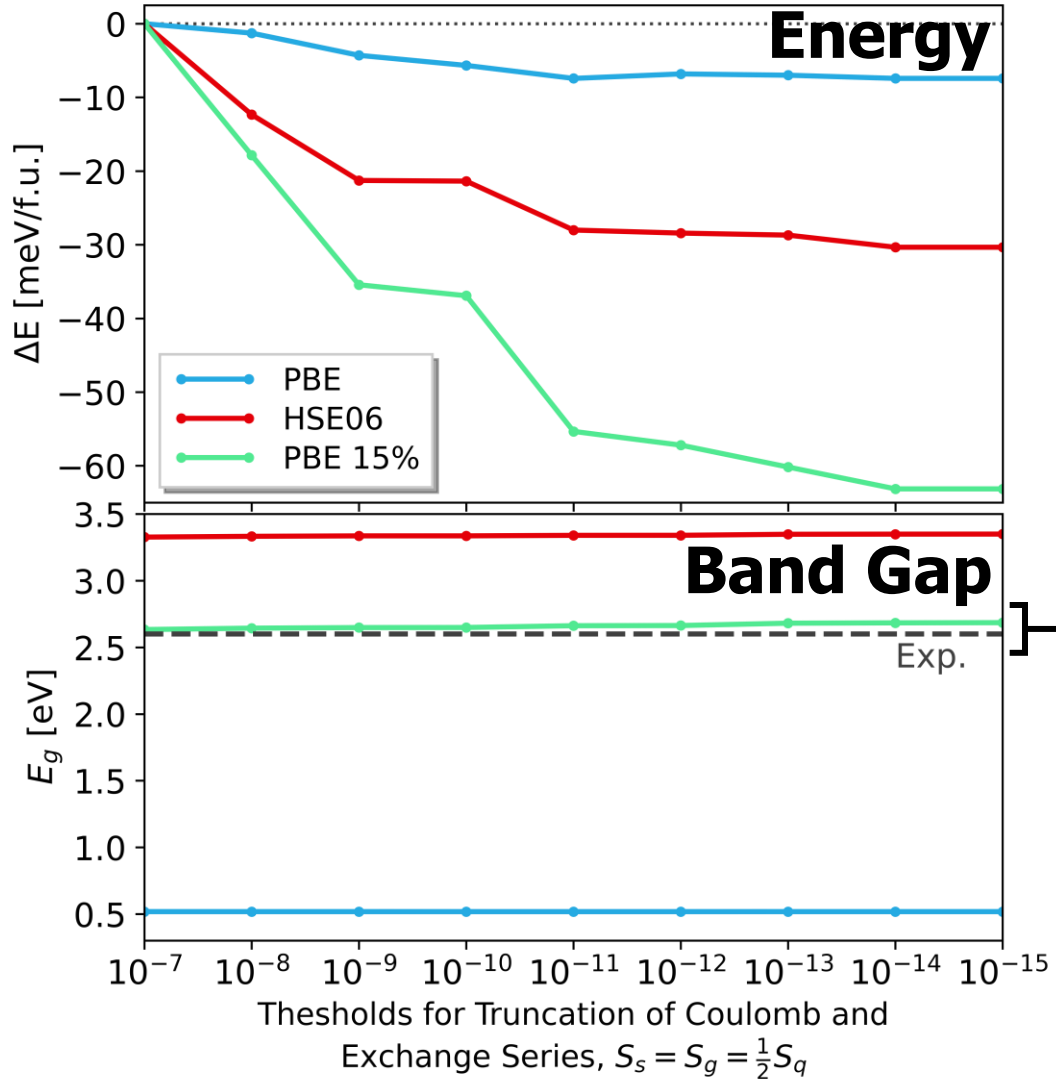
k point sampling in
12×12×12 Monkhorst-Pack mesh

All electron calculation

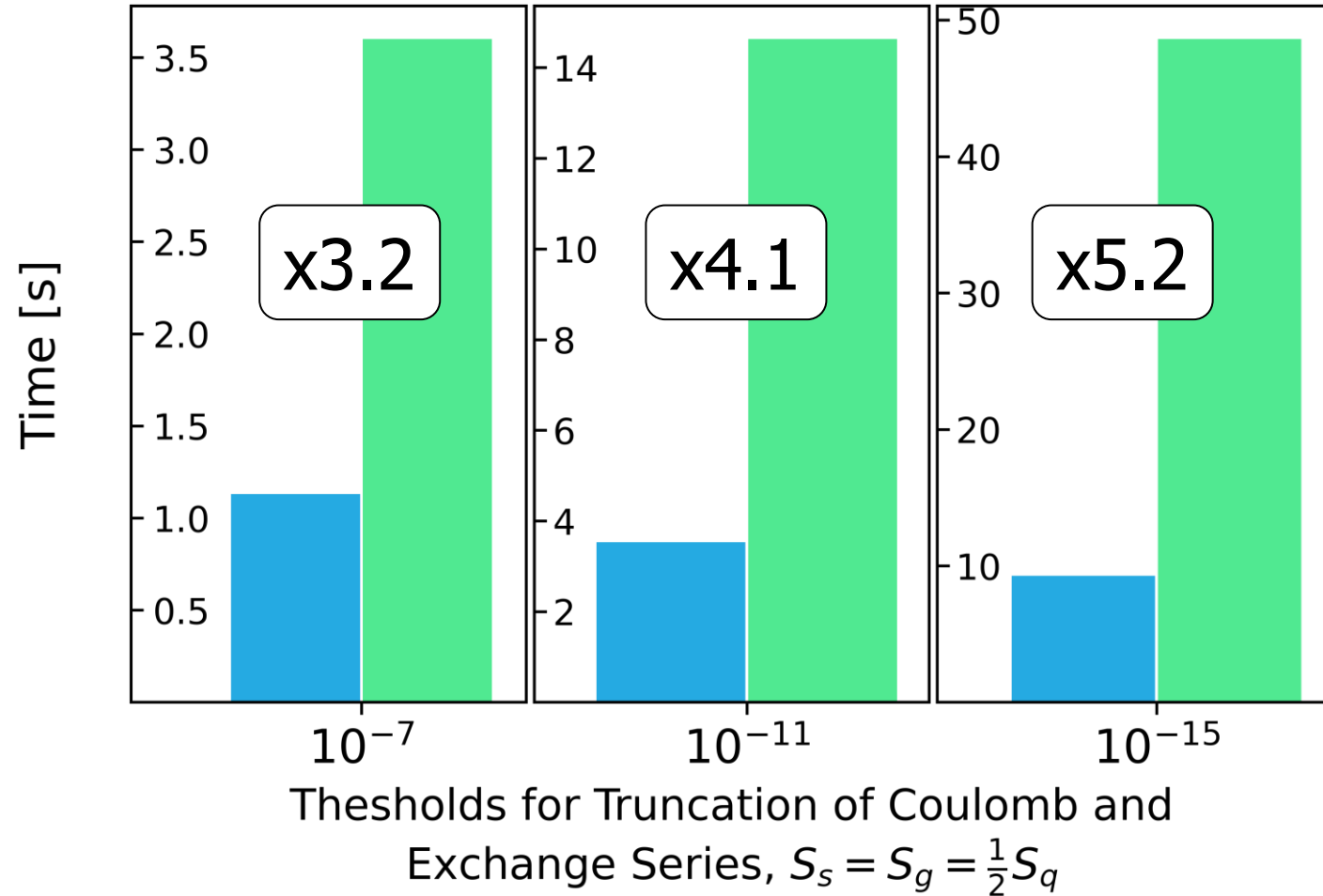
Hematite: Convergence of exx



Hematite: Convergence of exx



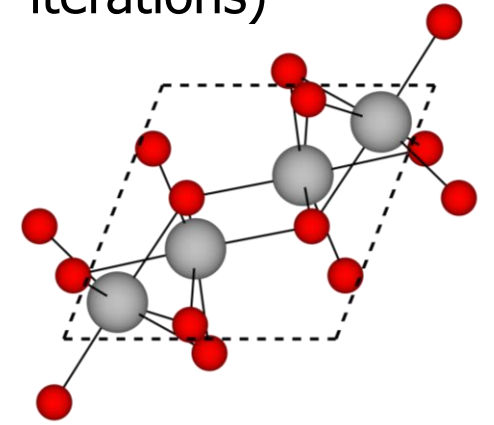
Hematite: Timings



PBE **PBE 15%**

Two-electron Integrals

Calculation
(avg on all SCF iterations)



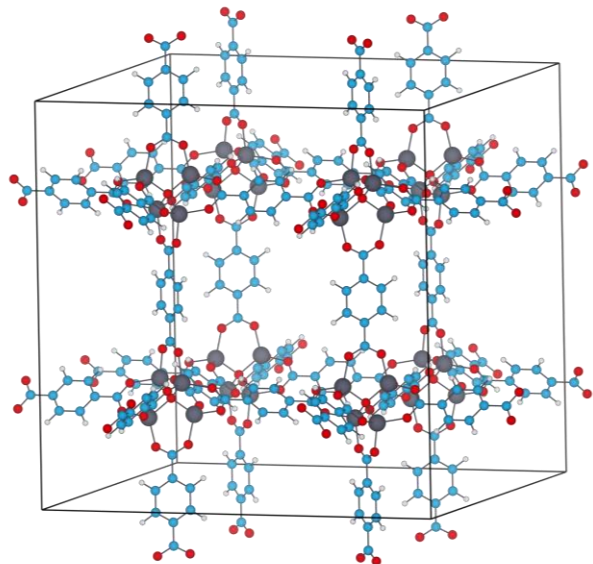
10 Atoms (two formula units)

186 AOs in Basis Set

1 Full Node of CPU Partition
of **Leonardo HPC**:

112 Cores (2x Intel Xeon 8480+)

Additional Test Cases

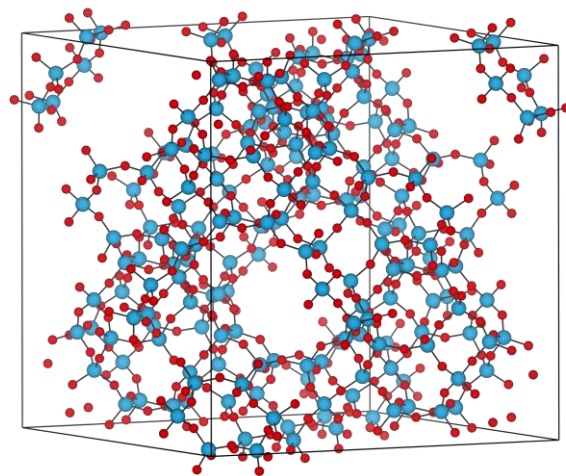


MOF-5

424 Atoms, 7184 AOs

Exploitation of
Symmetry

TIME [s]	PBE	B3LYP	
Two-elec. Integrals	1.84	4.39	x2.4
SCF Iteration	3.80	6.32	x1.66
Total	63.2	107.5	x1.7



FAUJASITE

576 Atoms, 11136 AOs

3 Nodes
60 CPU Cores
Intel Xeon
E5-2630

TIME [s]	PBE	B3LYP	
Two-elec. Integrals	24.36	39.72	x1.6
SCF Iteration	30.71	46.59	x1.5
Total	383.2	532.8	x1.4

$S_s = S_g = 7$
 $S_q = 15$

Acknowledgments

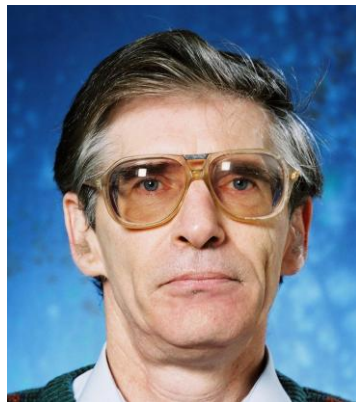
MSSC2025
TORINO 8-12 SEP 2025



Prof.
**Alessandro
Erba**



Prof.
**Jacques
Desmarais**



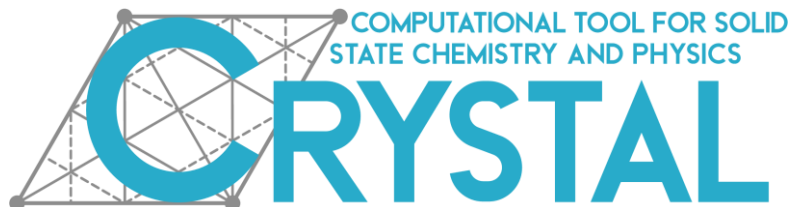
Dr.
**Vic
Saunders**



Theoretical Chemistry Group



**UNIVERSITÀ
DI TORINO**





UNIVERSITÀ
DI TORINO



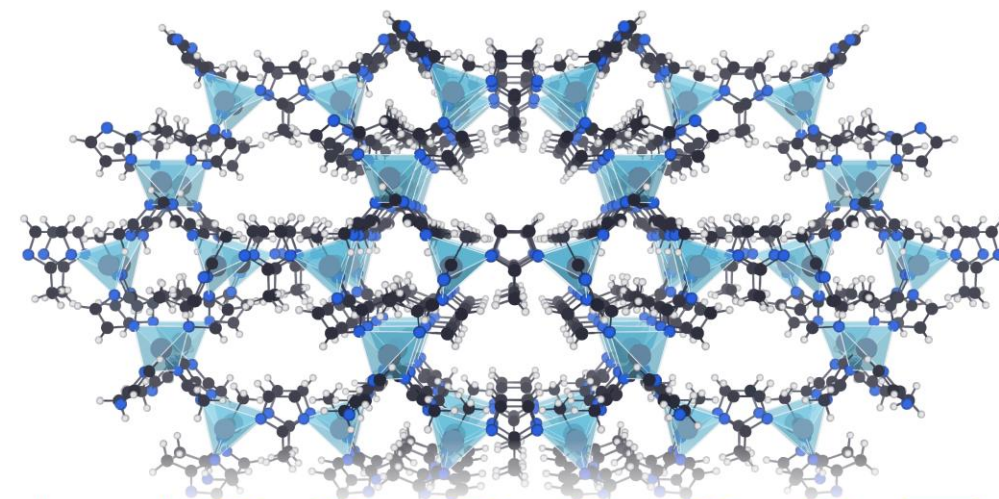
2023 - 2027

DIPARTIMENTO
DI ECCELLENZA

Ministero dell'Università e della Ricerca



 CRYSTAL23



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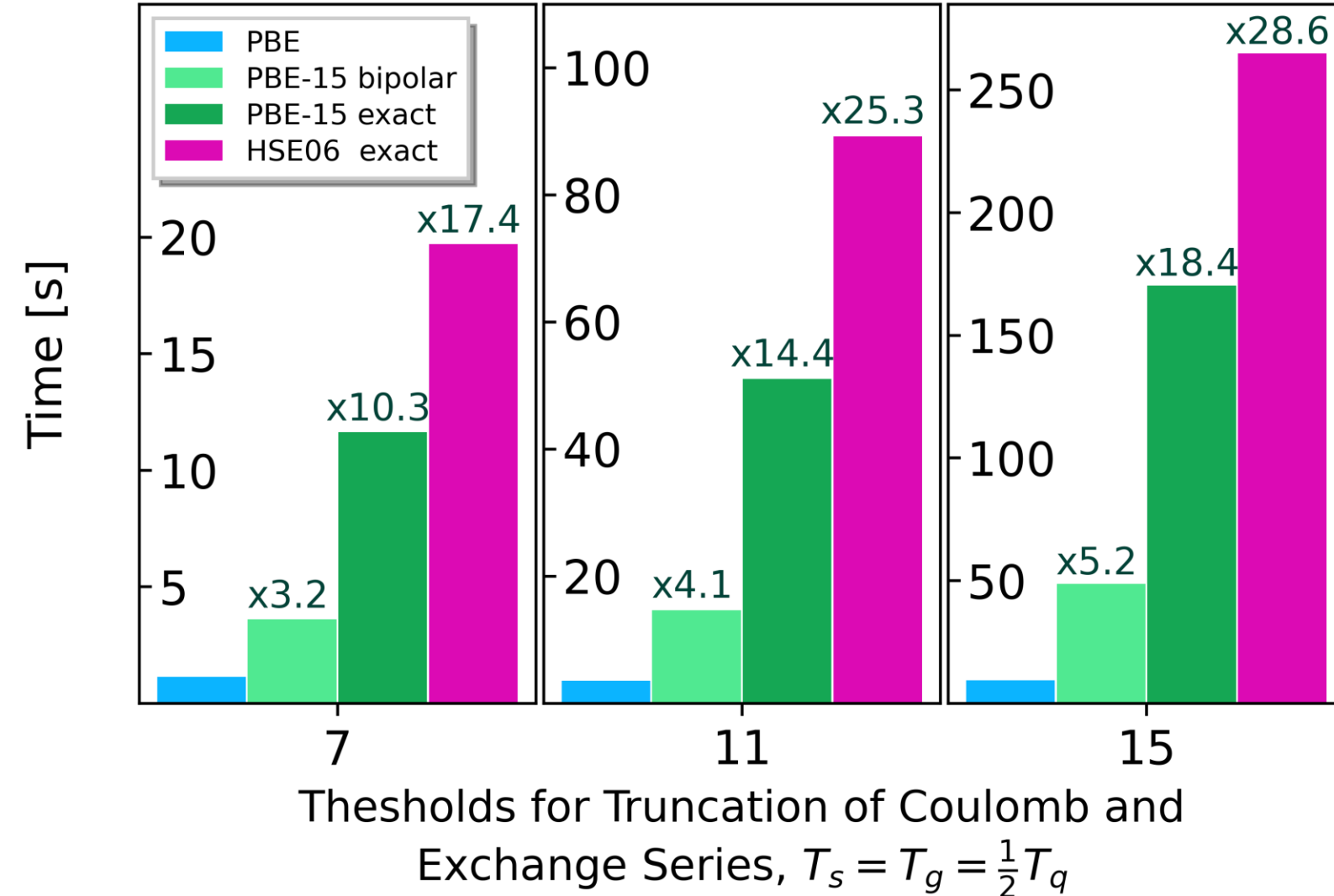
Thanks for Your Attention



Finanziato
dall'Unione europea
NextGenerationEU

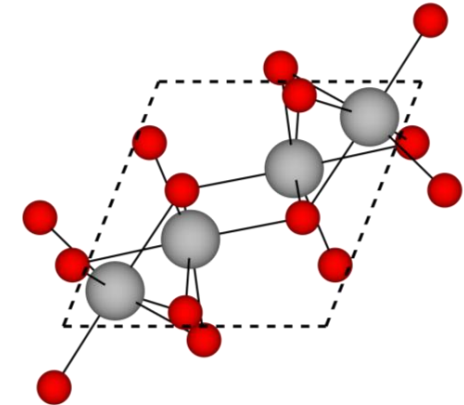
 ICSC
Centro Nazionale di Ricerca in HPC,
Big Data and Quantum Computing

Hematite: Timings



Two-electron Integrals

Calculation
(avg on all SCF iterations)



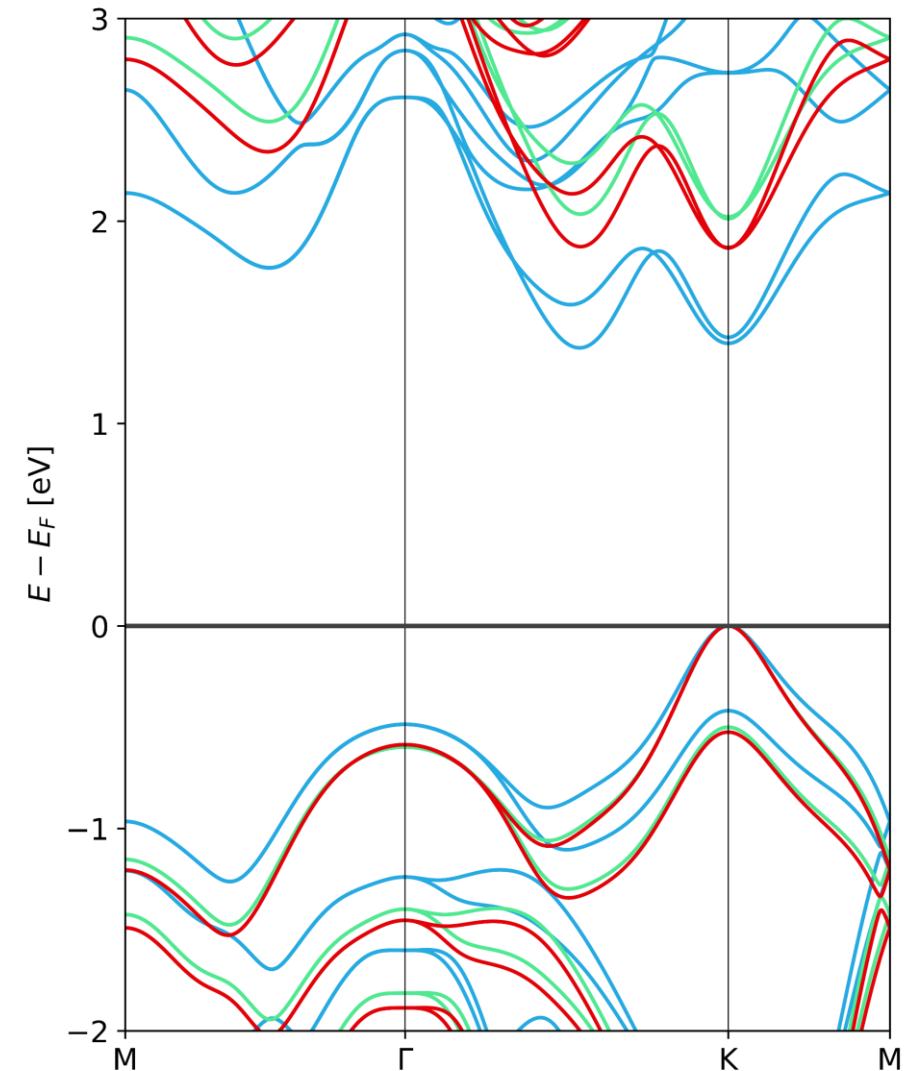
10 Atoms (two formula units)

186 AOs in Basis Set

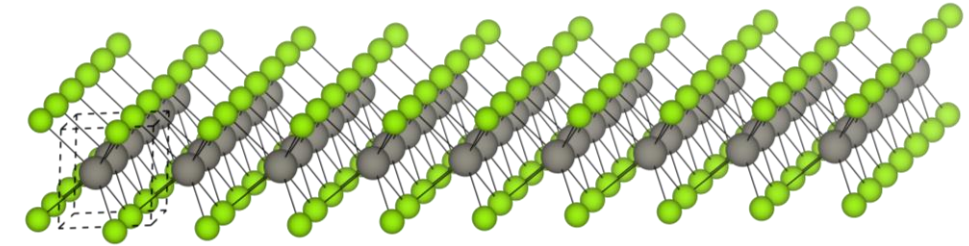
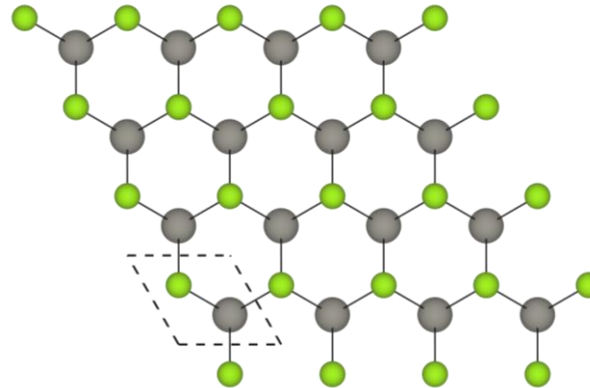
1 Full Node of CPU Partition
of **Leonardo HPC**:

112 Cores (2x Intel Xeon 8480+)

WSe₂: Electronic Structure



HSE06
PBE 15%
PBE

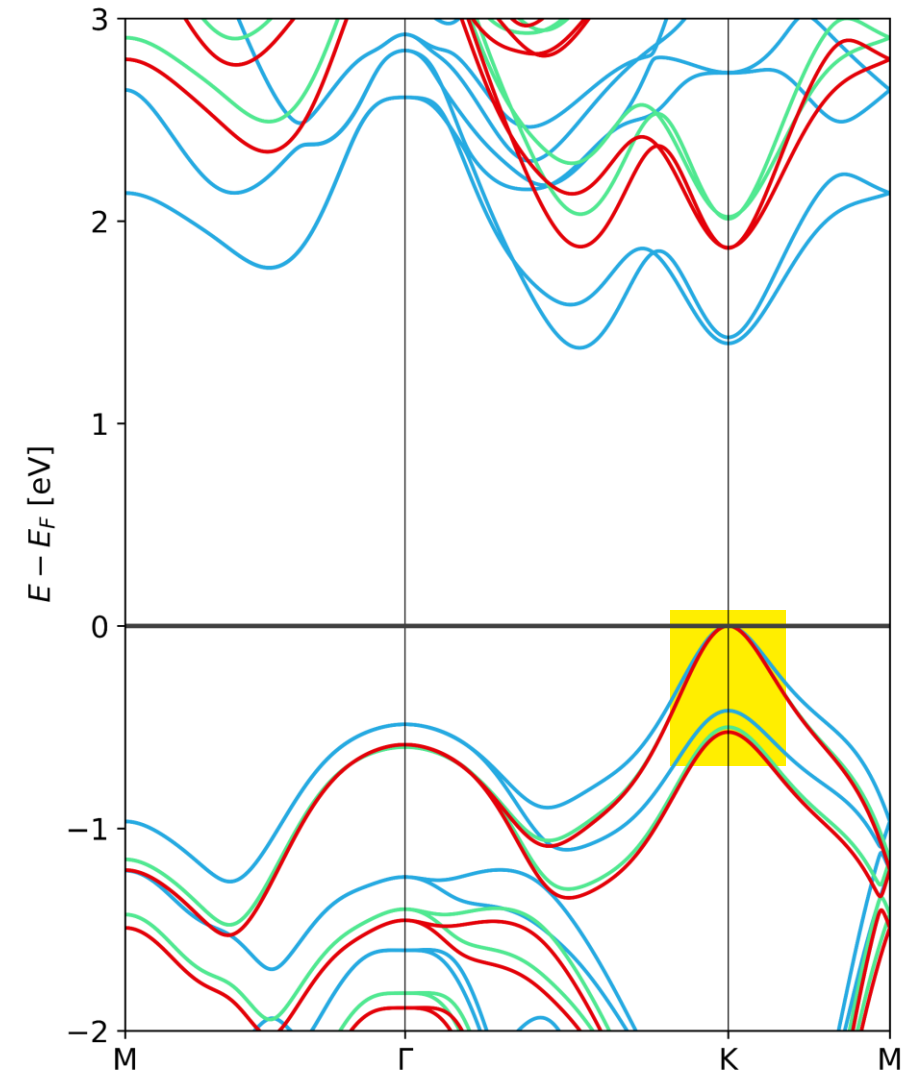


2D System

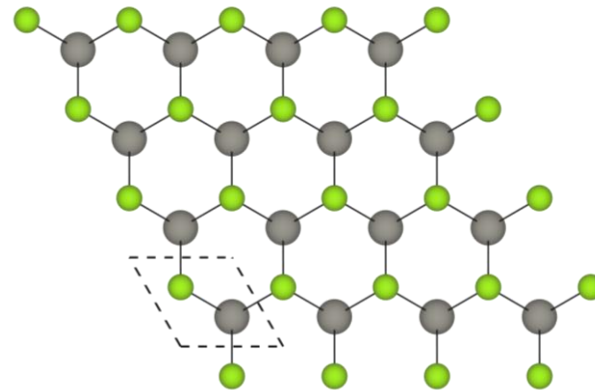
Inclusion of relativistic effects
(SOC) with core pseudopotentials

Space Inversion Symmetry Breaking
Time-Reversal Symmetry Preservation
Large Spin-Orbit Coupling

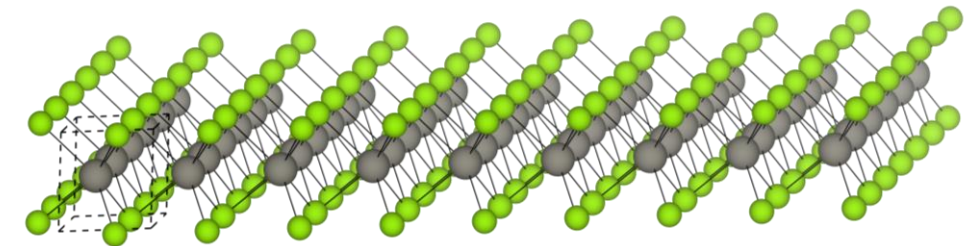
WSe₂: Electronic Structure



HSE06
PBE 15%
PBE



**Rashba type-I
Spin Splitting**



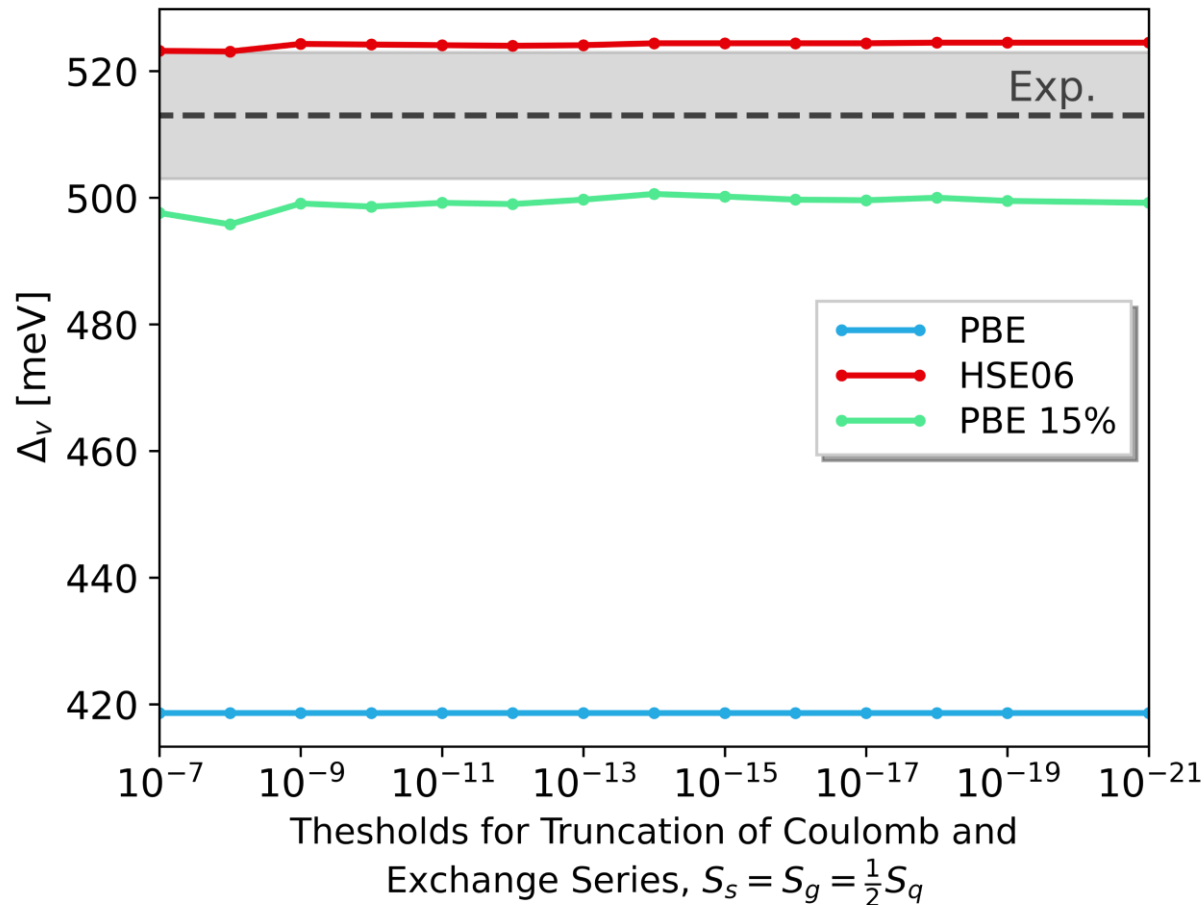
2D System

Inclusion of relativistic effects
(SOC) with core pseudopotentials

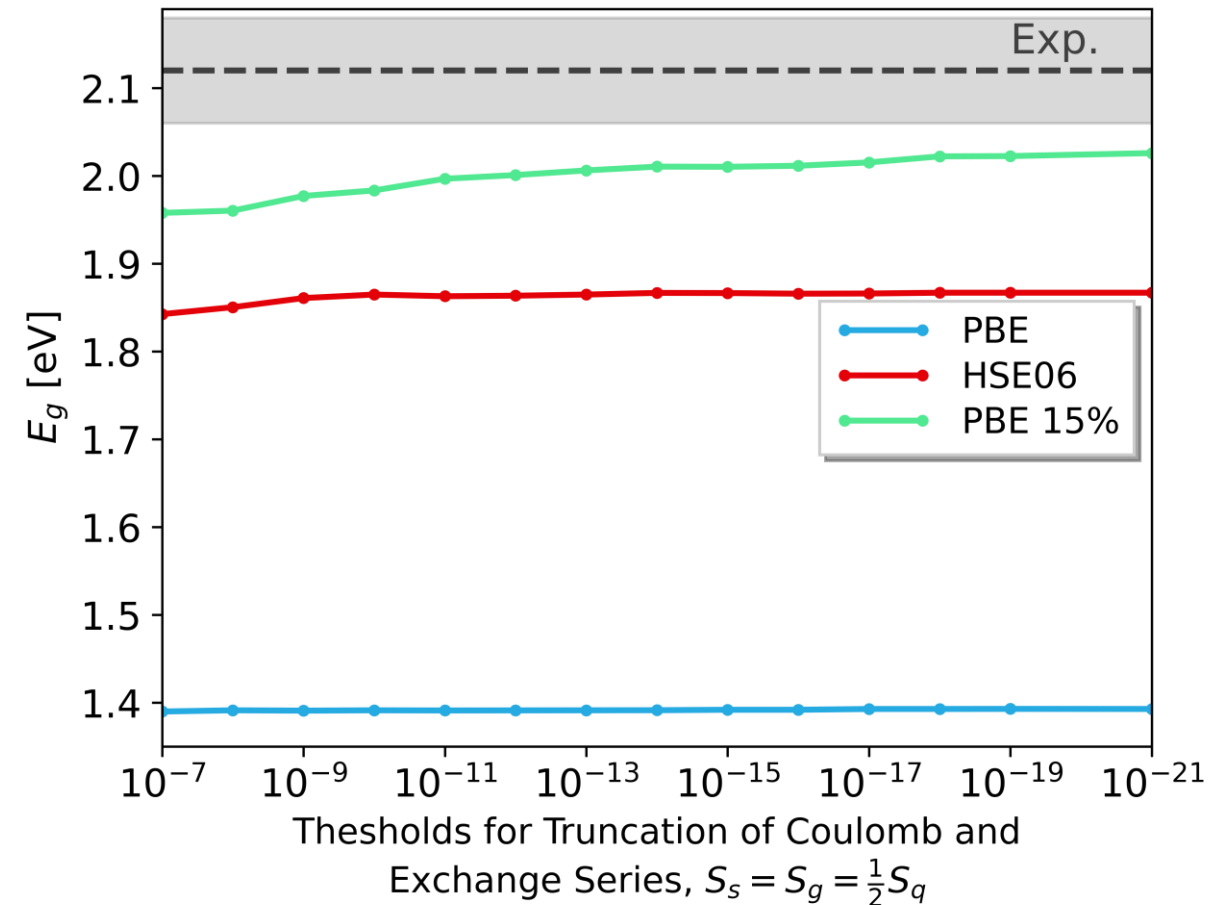
Space Inversion Symmetry Breaking
Time-Reversal Symmetry Preservation
Large Spin-Orbit Coupling

WSe₂: Convergence of exx

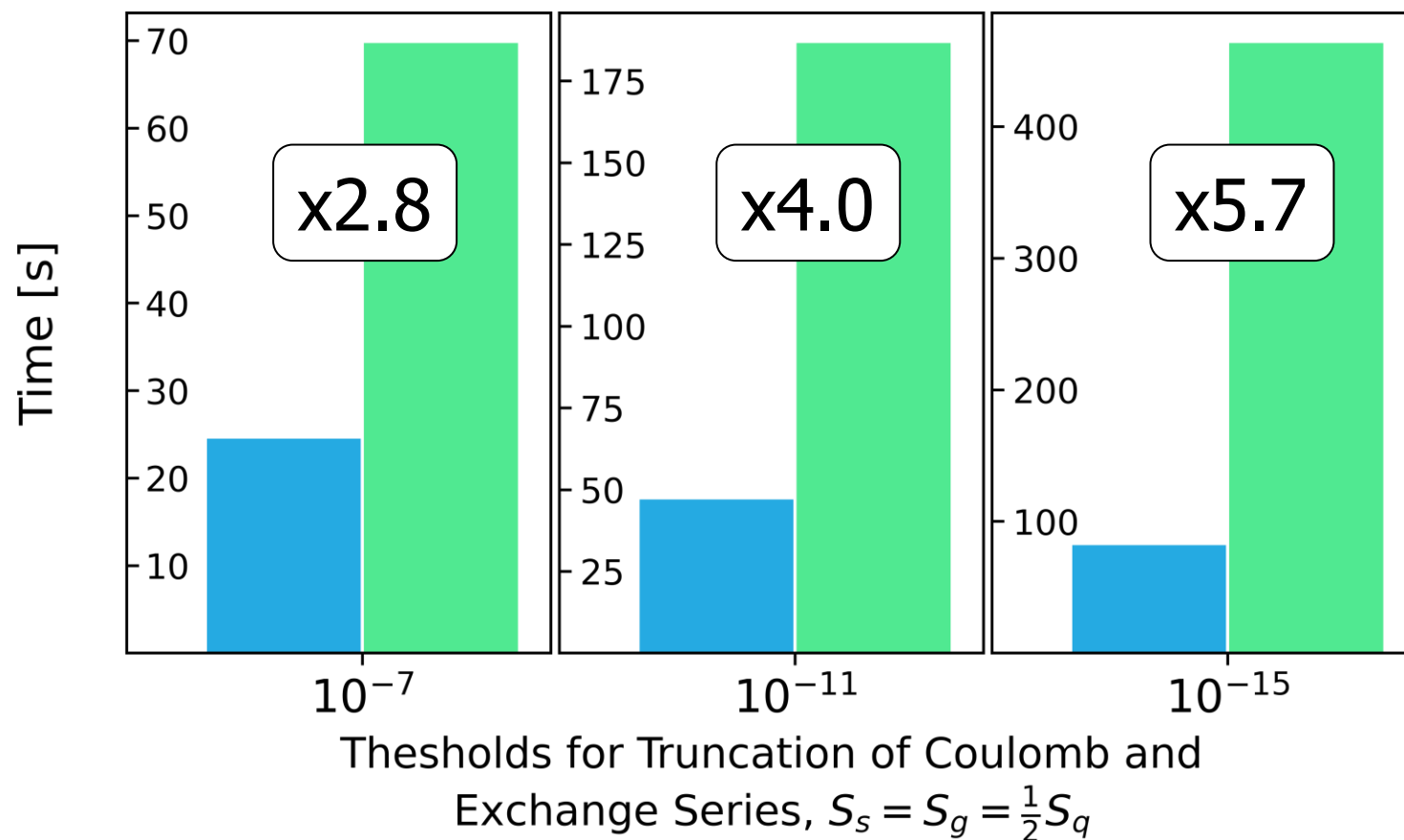
Rashba Splitting



Band Gap

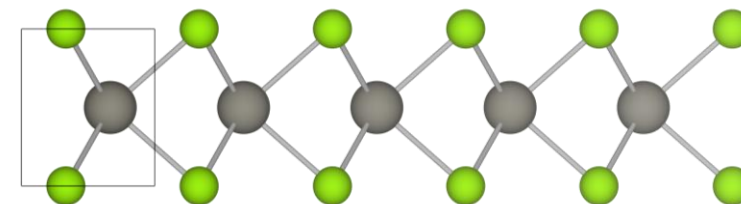


WSe₂: Timings



PBE **PBE 15%**

Full SCF Iteration (avg)



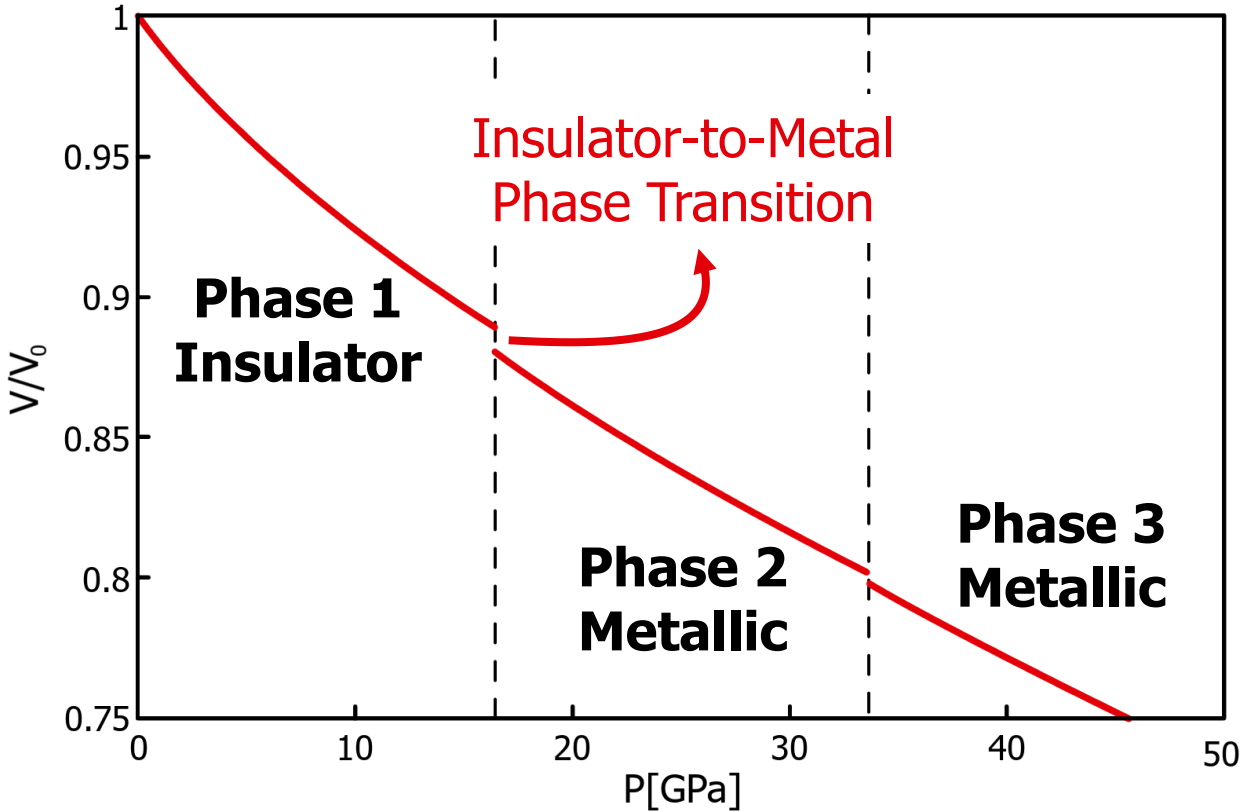
3 Atoms

125 AOs in Basis Set
(+ Pseudopotentials)

1 Full Node of CPU Partition
of **Leonardo HPC**:

112 Cores (2x Intel Xeon 8480+)

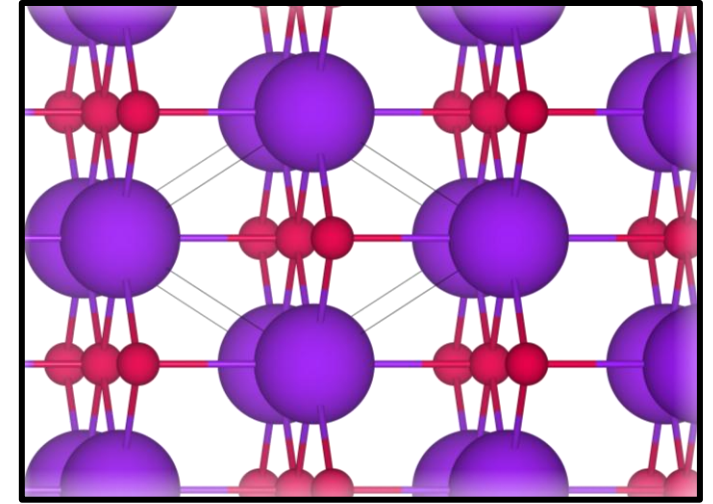
EuO: Structure



Stress Test for our Approach:

High Density, Metallic Character

Very Slow Decay of the Exact Exchange Series



Under Pressure: 20 GPa (High Density)

Suitable for Hybrid DFT:

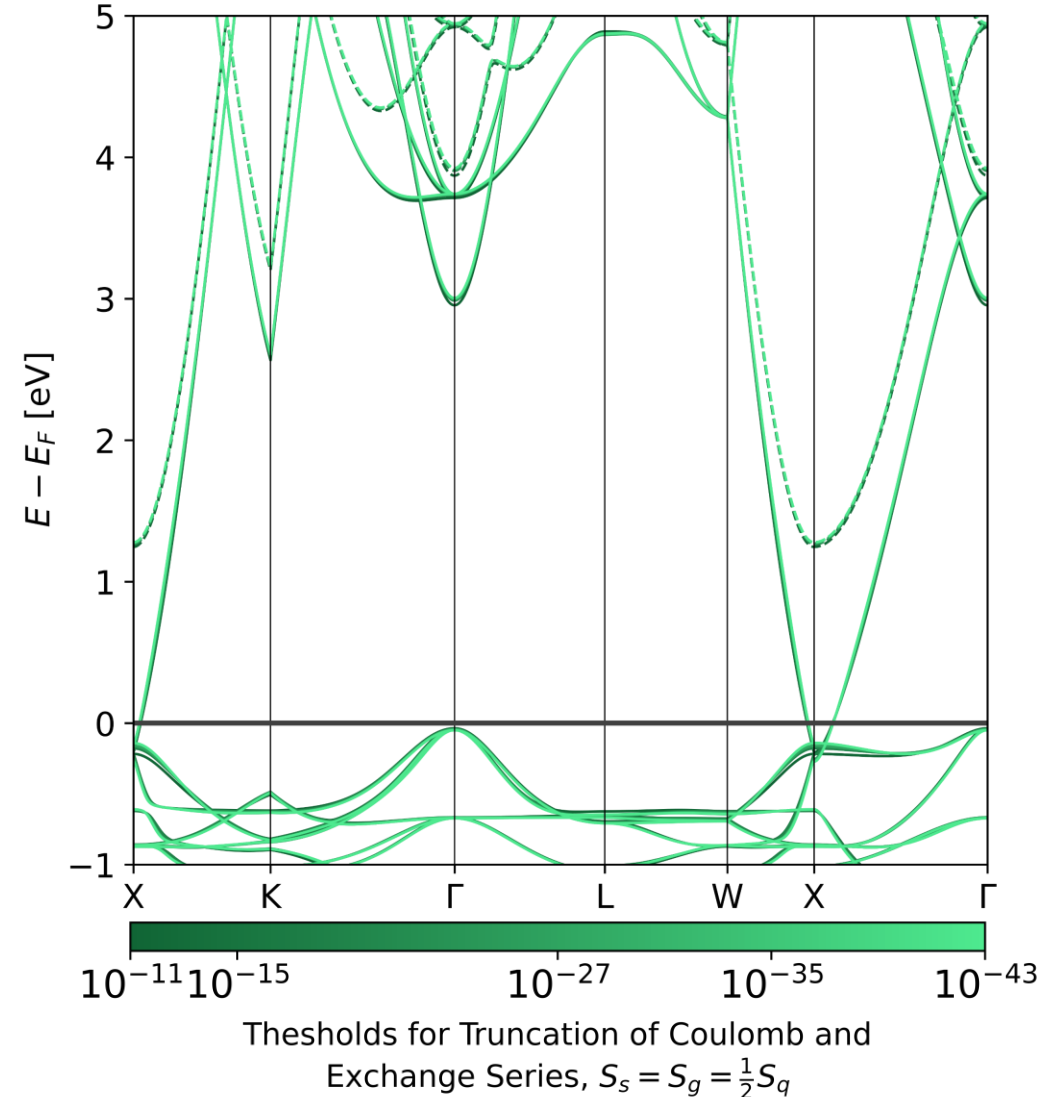
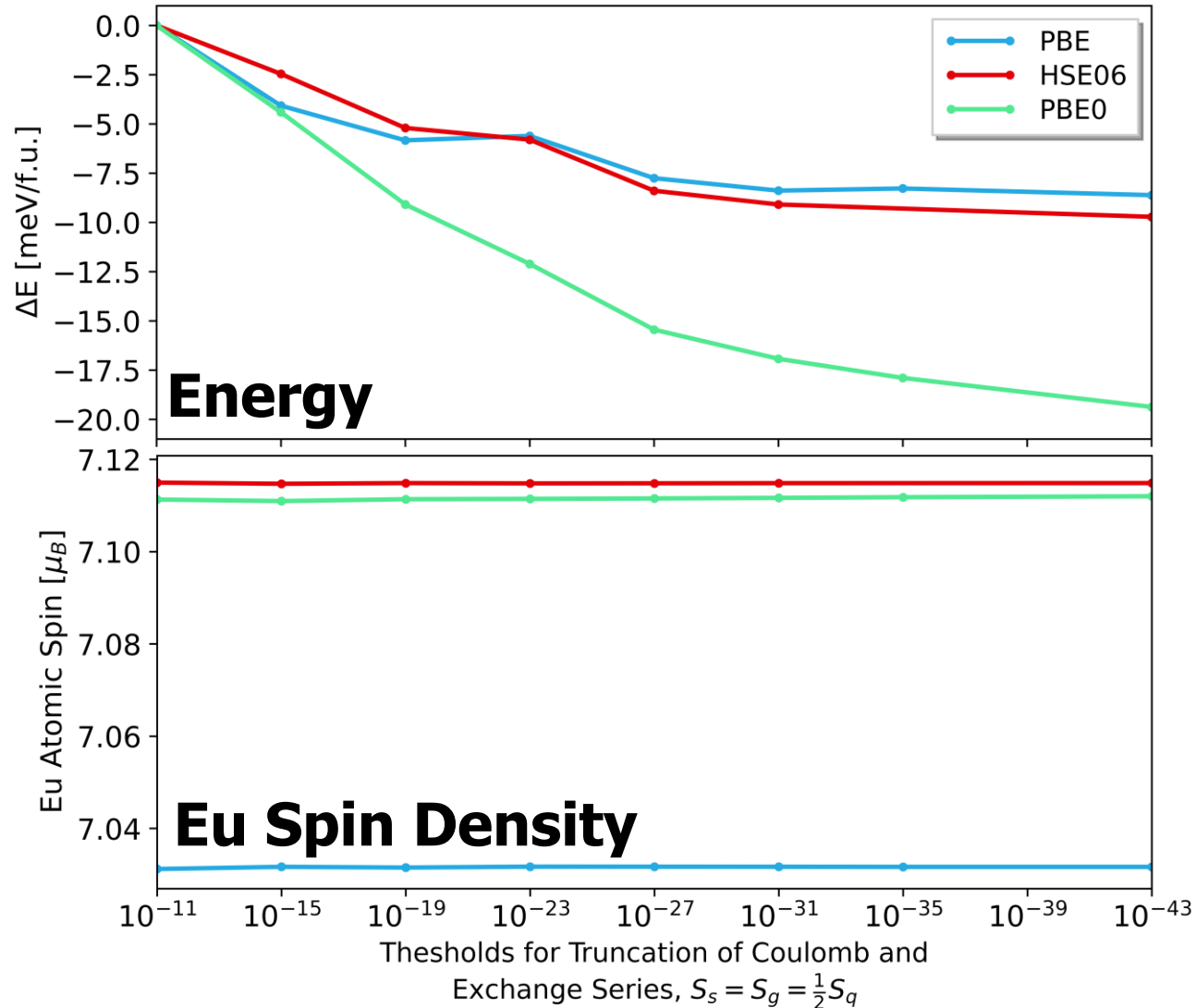
7 Unpaired Electrons in 4f Orbital on each Eu

Highly Localized Electrons

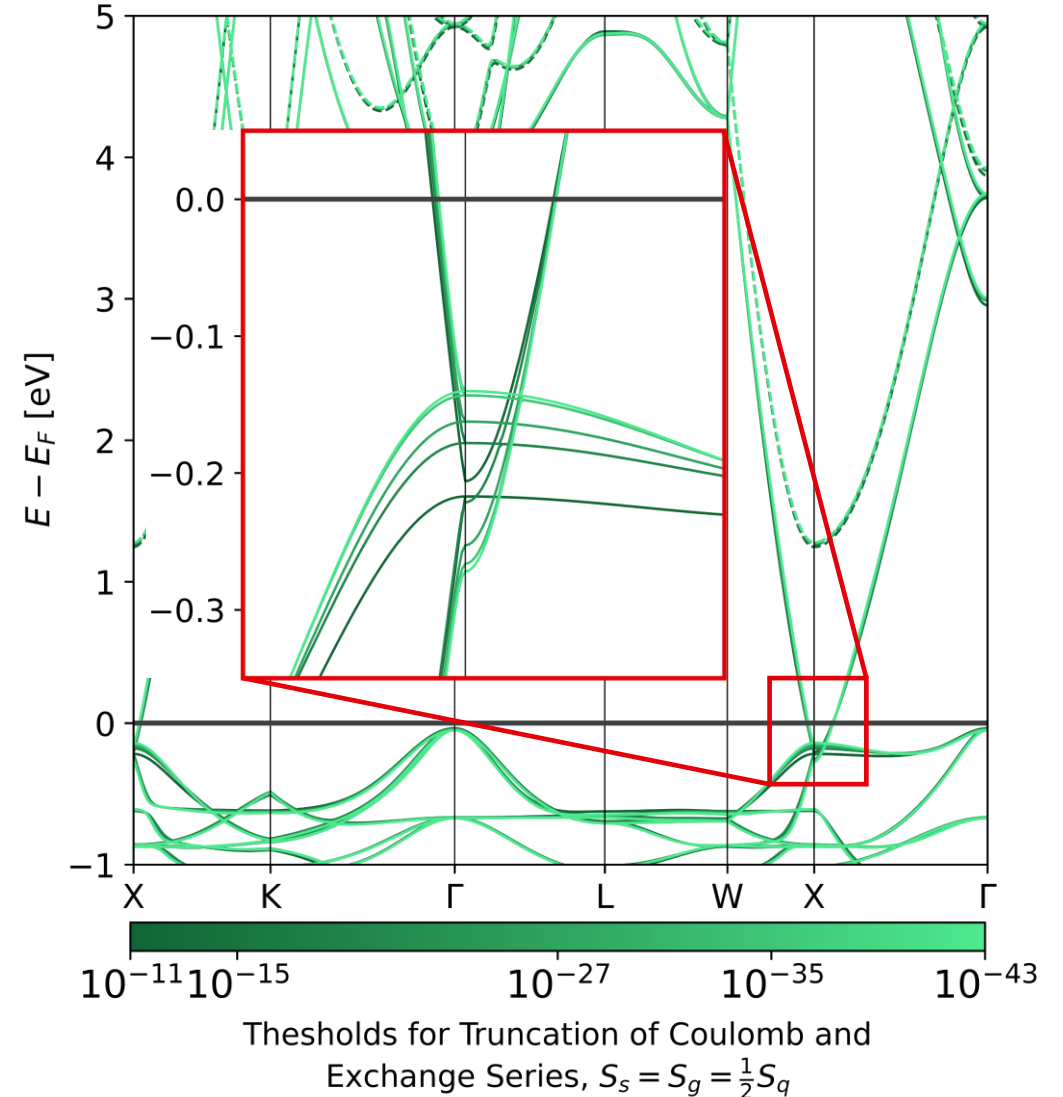
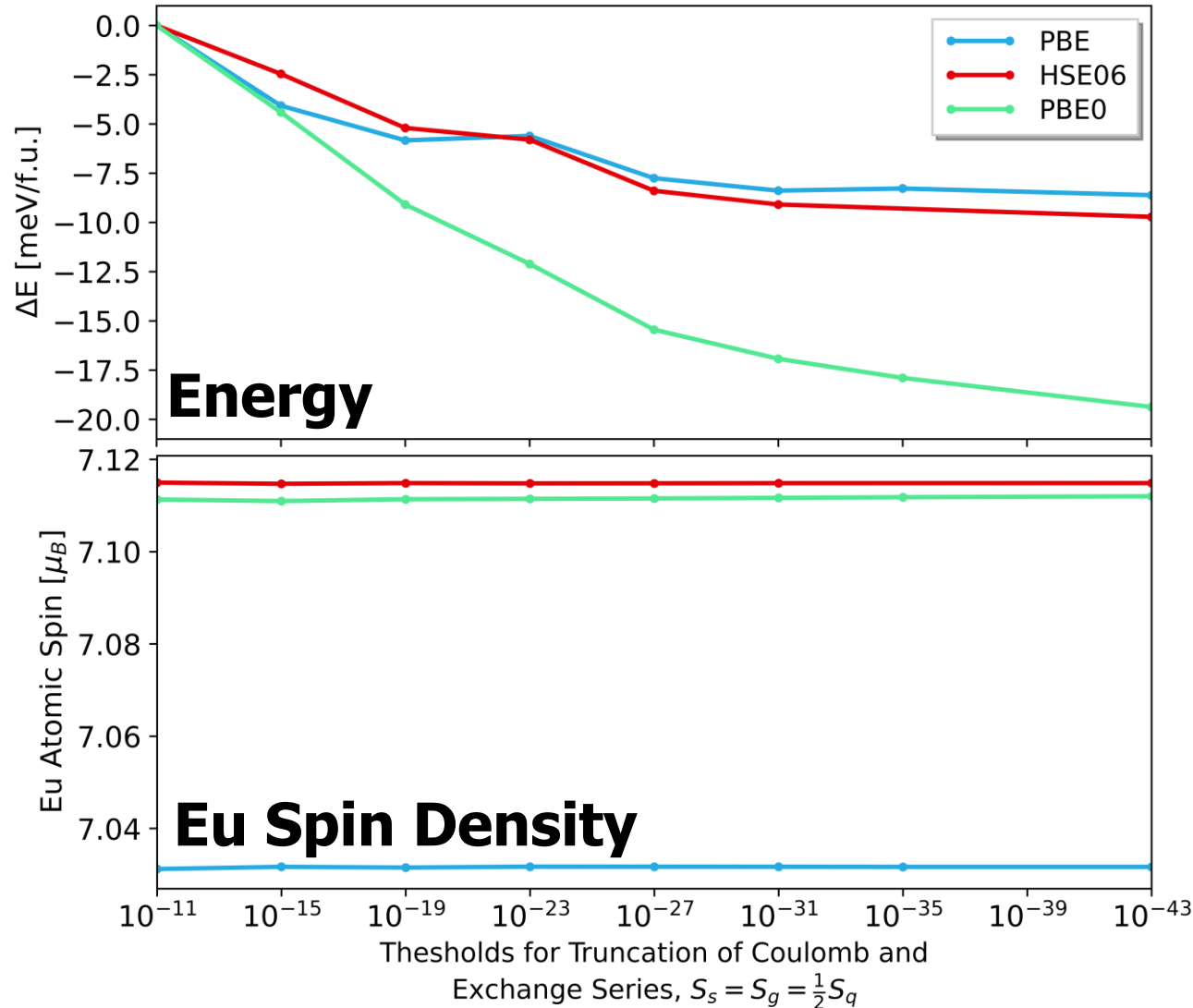
Metallic

Highly Delocalized Electrons

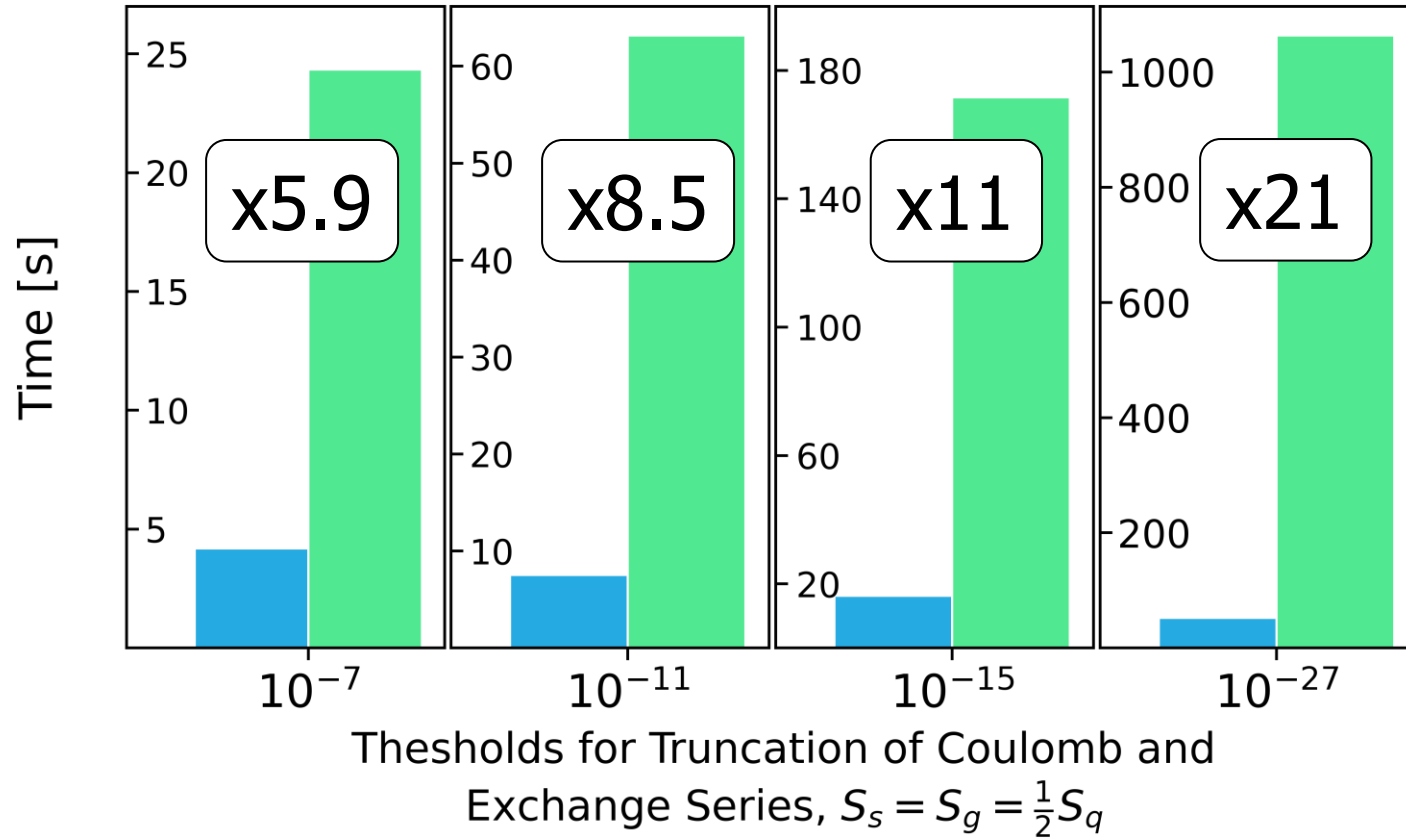
EuO: Electronic Structure



EuO: Electronic Structure

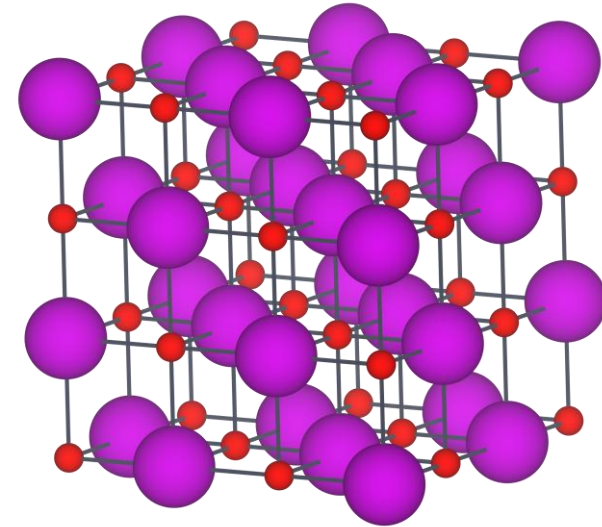


EuO: Timings



PBE **PBE 15%**

Full SCF Iteration (avg)



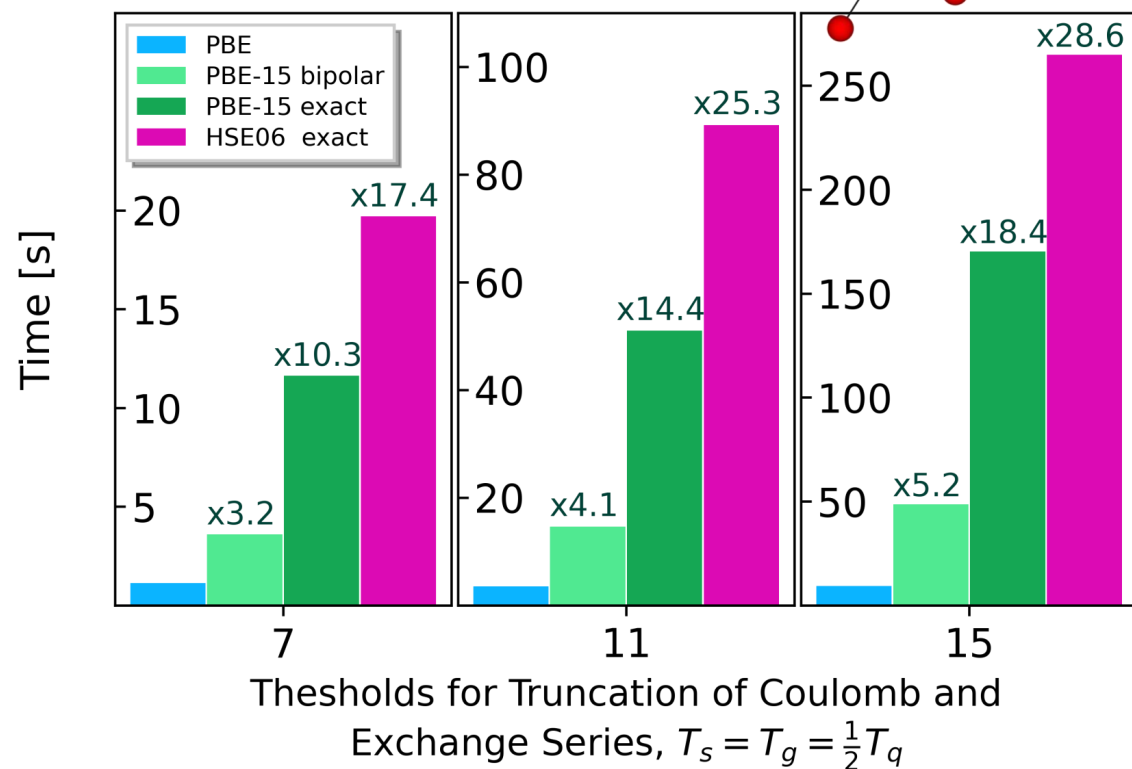
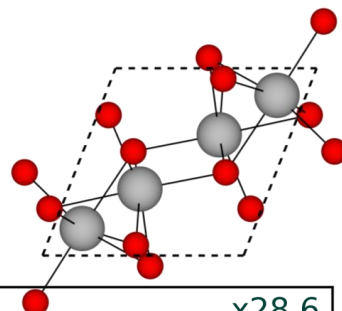
2 Atoms

147 AOs in Basis Set
(+ Pseudopotential for Eu)

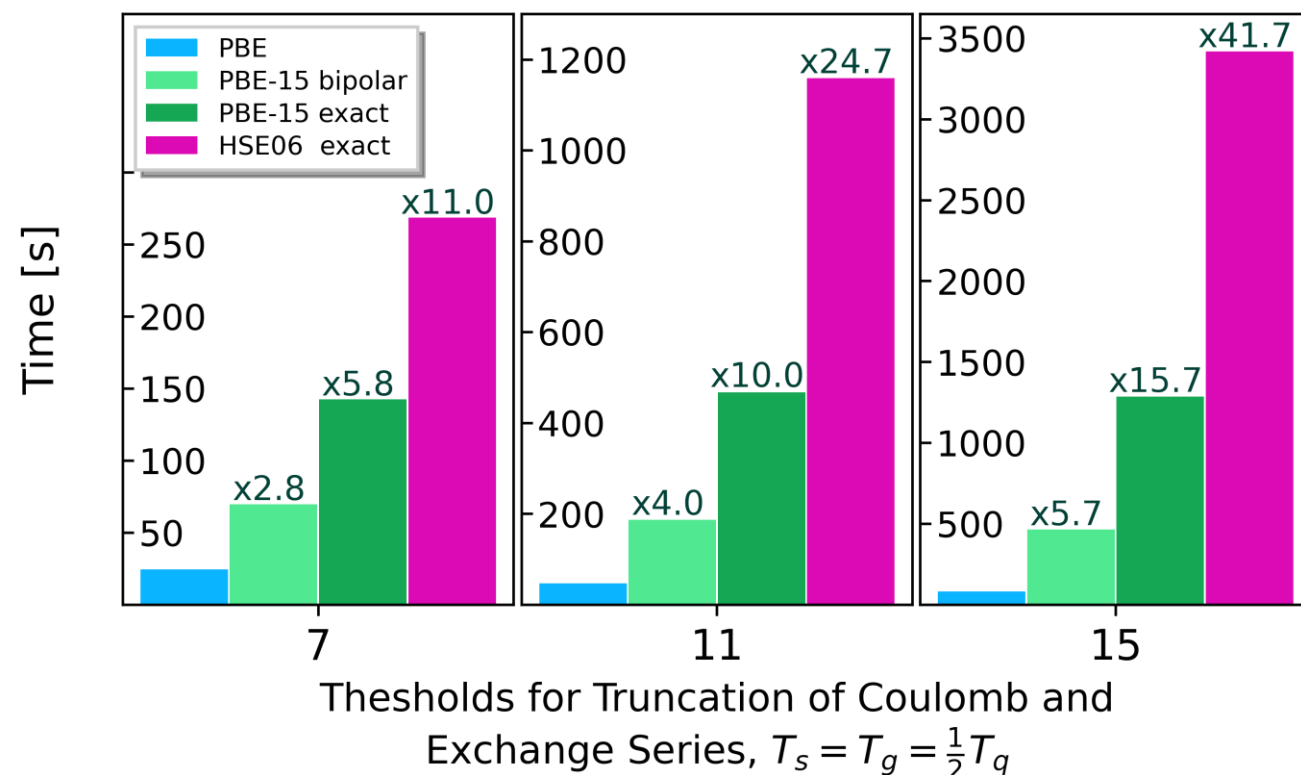
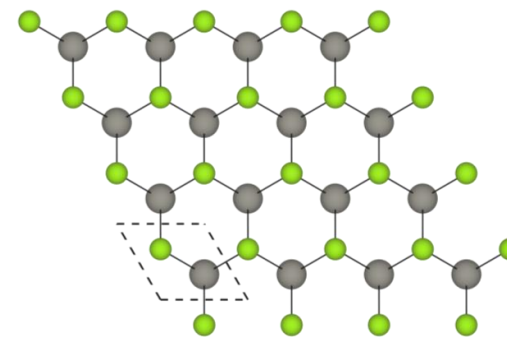
1 Full Node of CPU Partition
of **Leonardo HPC**:
112 Cores (2x Intel Xeon 8480+)

Full Timings

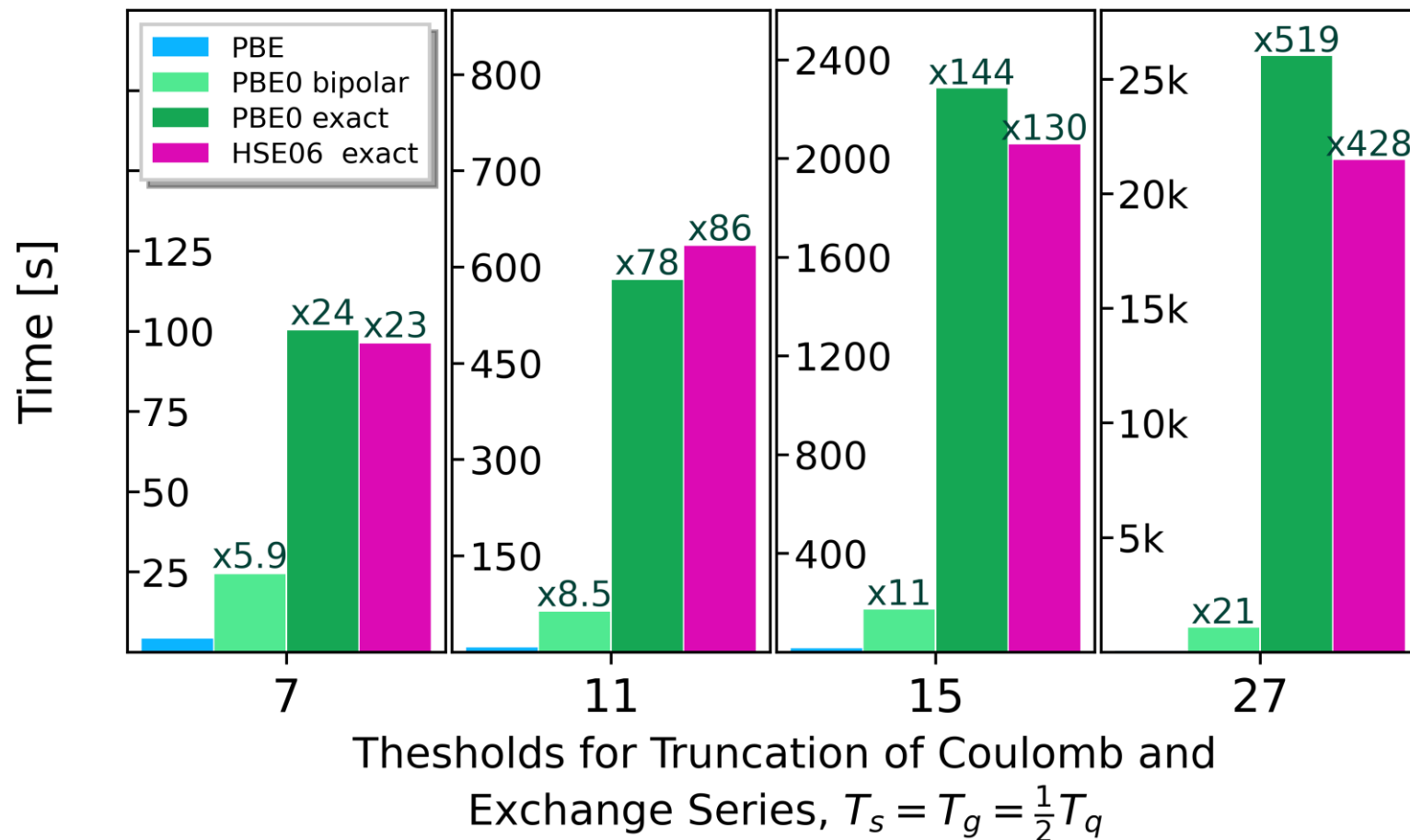
Two-electron Integrals Calculation (avg on all SCF iterations)



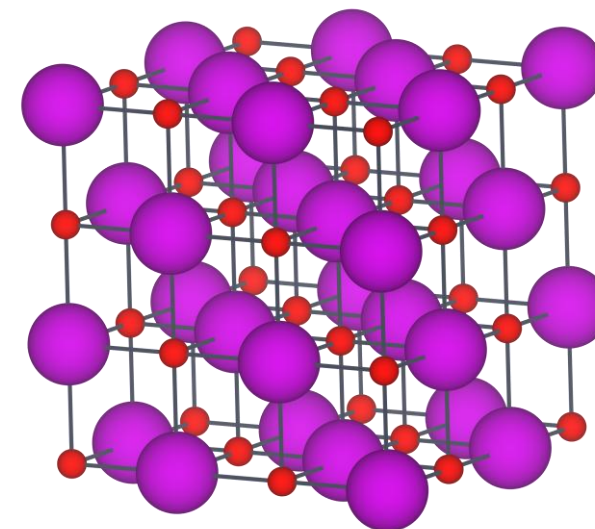
Full SCF Iteration (avg)



Full Timings



Full SCF Iteration (avg)



Bipolar Expansion

$$(\chi_\mu^0 \chi_\nu^s | \chi_\sigma^g \chi_\rho^{s+q}) = \iint d\mathbf{r}_1 d\mathbf{r}_2 \frac{\chi_\mu^0(\mathbf{r}_1) \chi_\nu^s(\mathbf{r}_1) \chi_\sigma^g(\mathbf{r}_2) \chi_\rho^{s+q}(\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|}$$

$$\frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|}$$

Bipolar Expansion in
terms of **Spherical**
Harmonics

$$\frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} = \sum_{l_1, l_2} \frac{1}{R^{2l+1}} \sum_{m_1, m_2} B(l_1, m_1, l_2, m_2) Y_l^{-m}(\mathbf{s}_2; \mathbf{s}_1) Y_{l_1}^{m_1}(\mathbf{s}_1; \mathbf{r}_1) Y_{l_2}^{m_2}(\mathbf{s}_2; \mathbf{r}_2)$$

$$\left[\begin{aligned} \rho_1(\mathbf{s}_1; \mathbf{r}_1) &= \chi_\mu(\mathbf{r}_1) \chi_\nu(\mathbf{r}_1) \\ \rho_2(\mathbf{s}_2; \mathbf{r}_2) &= \chi_\sigma(\mathbf{r}_2) \chi_\rho(\mathbf{r}_2) \end{aligned} \right.$$

For Well-Separated
Non-Overlapping Distributions

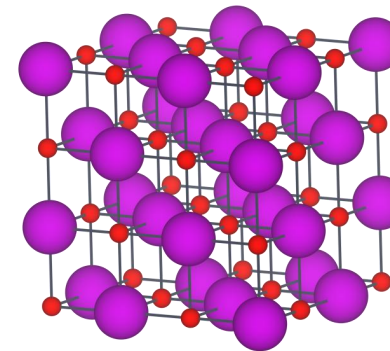
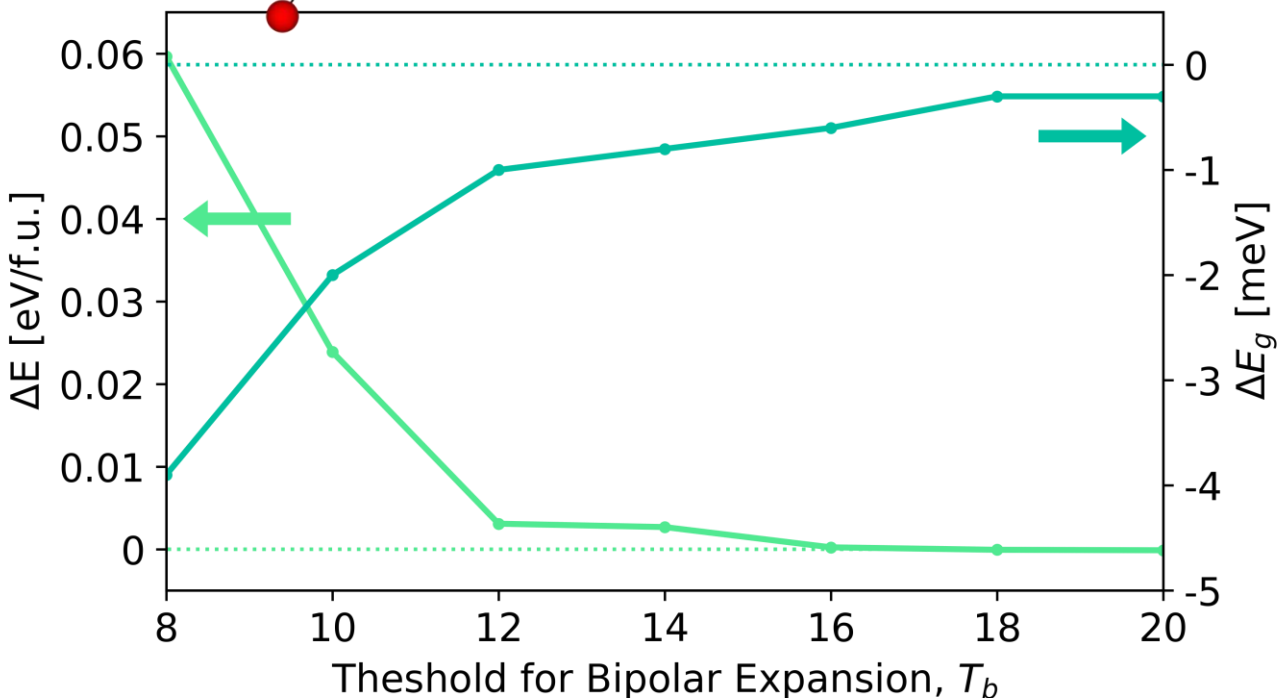
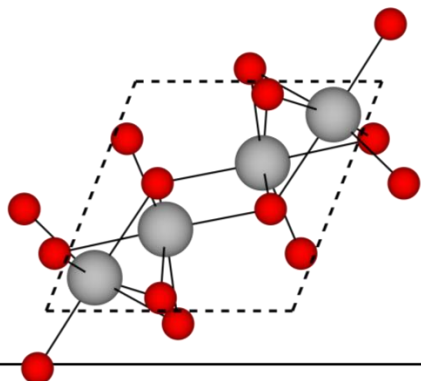
Overlap Based
Criterion

$$S_b$$

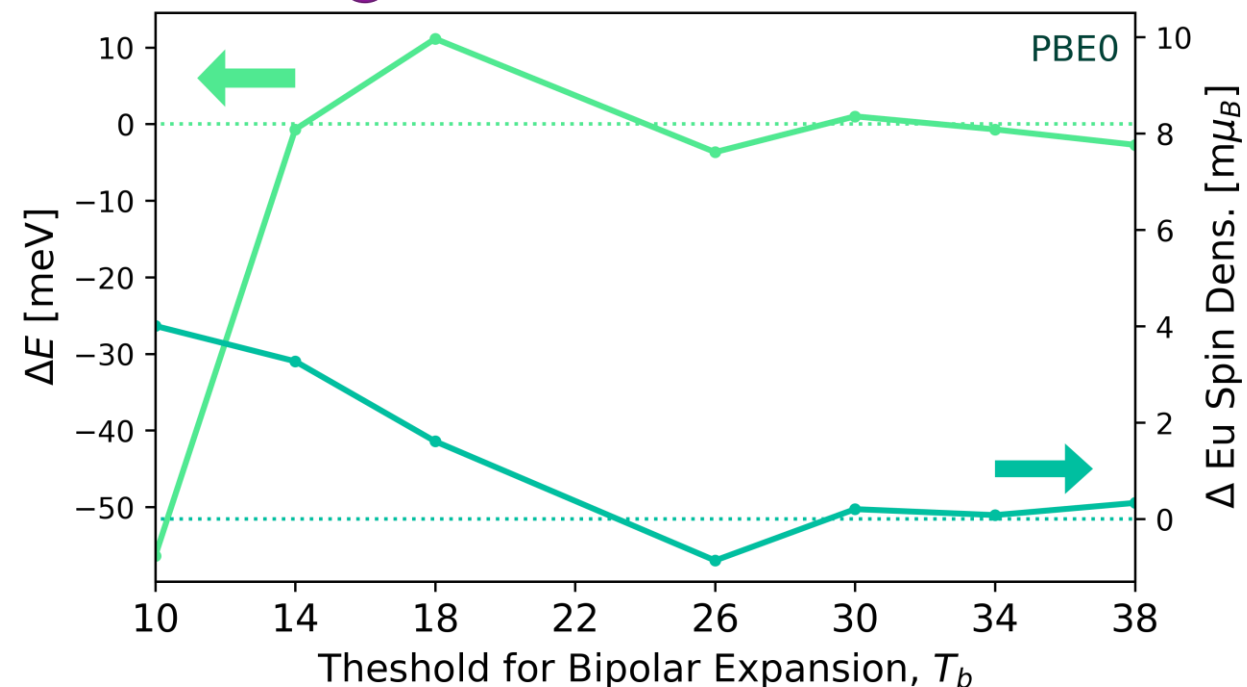
$$(\chi_\mu \chi_\nu | \chi_\sigma \chi_\rho) = \sum_{l_1, l_2} \sum_{m_1, m_2} B(l_1, m_1, l_2, m_2) \frac{Y_l^{-m}(\mathbf{s}_2; \mathbf{s}_1)}{R^{2l+1}} \left[\int d\mathbf{r}_1 \rho_1(\mathbf{s}_1; \mathbf{r}_1) Y_{l_1}^{m_1}(\mathbf{s}_1; \mathbf{r}_1) \right] \left[\int d\mathbf{r}_2 \rho_2(\mathbf{s}_2; \mathbf{r}_2) Y_{l_2}^{m_2}(\mathbf{s}_2; \mathbf{r}_2) \right]$$

Double Integral $\rightarrow$ Product of Single Integrals (Reduced Computational Cost)

Bipolar Expansion: Convergence



14 is the default
value for T_b

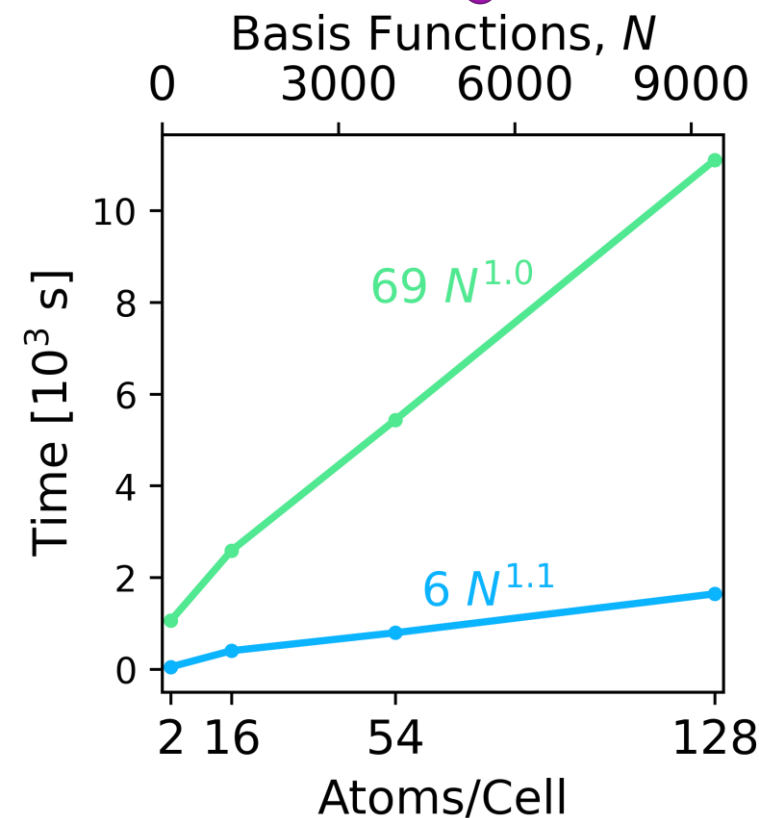
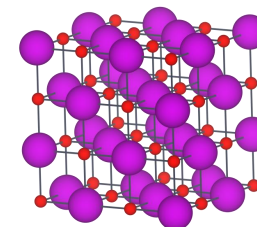
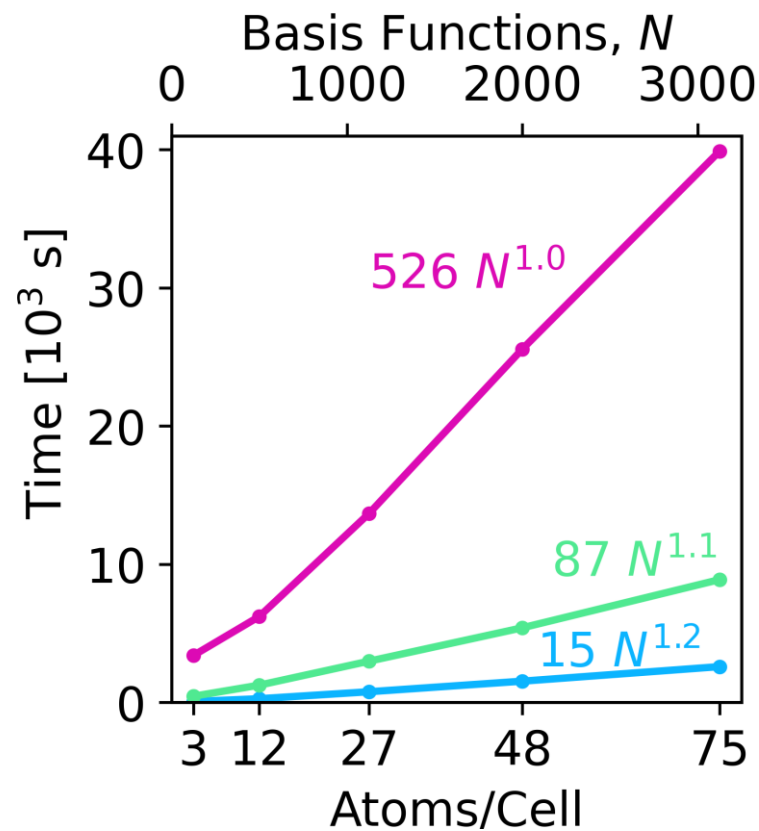
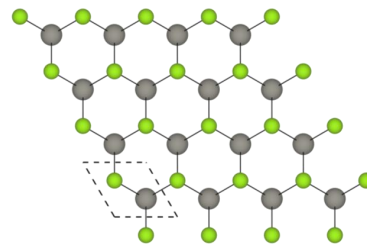
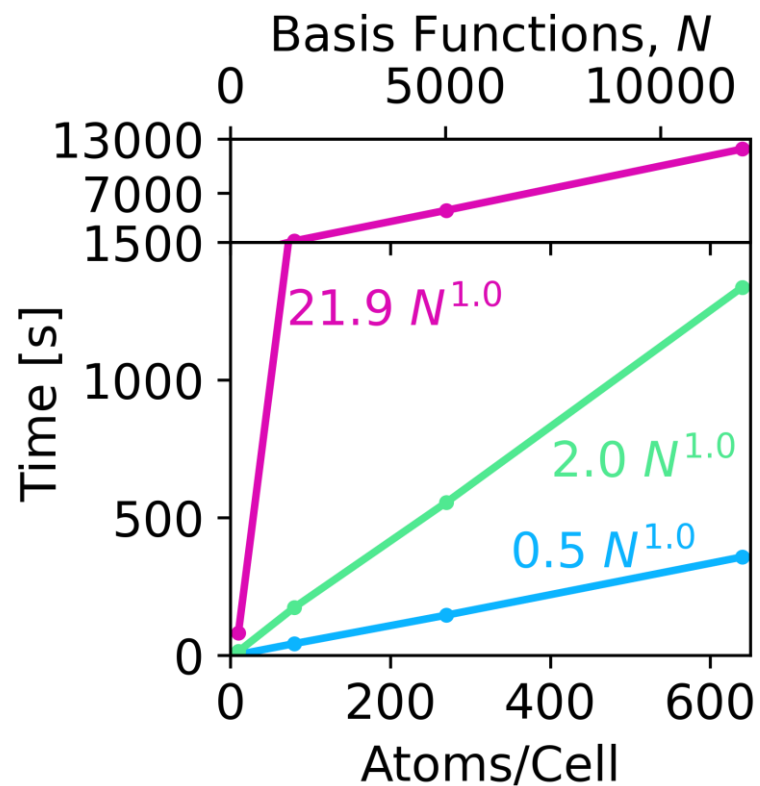
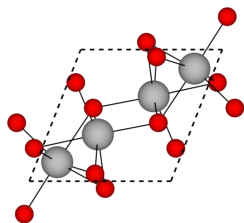


Weak Scaling

HSE06

PBE - 15%

PBE

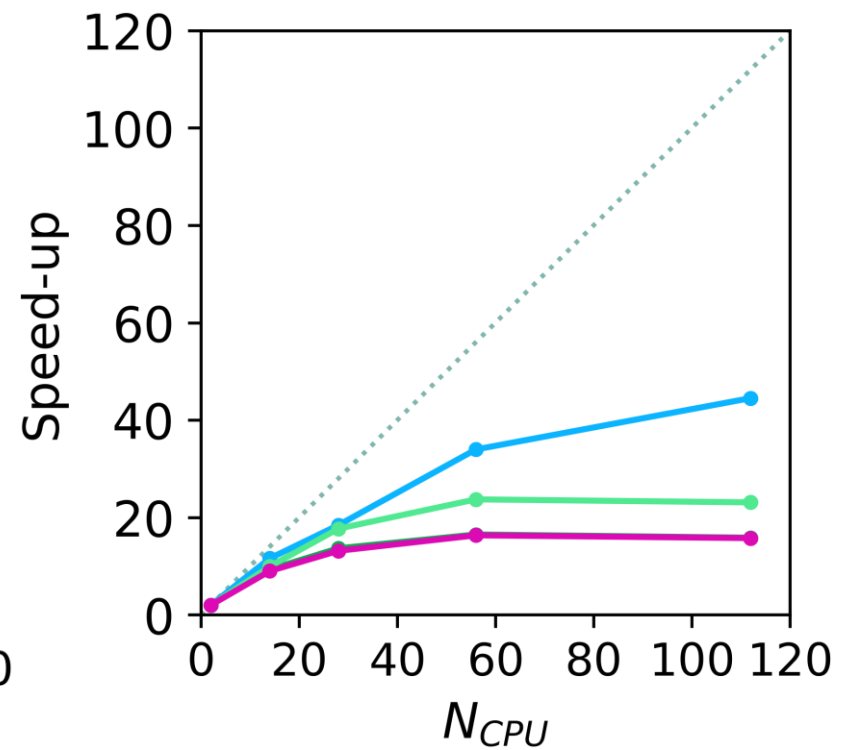
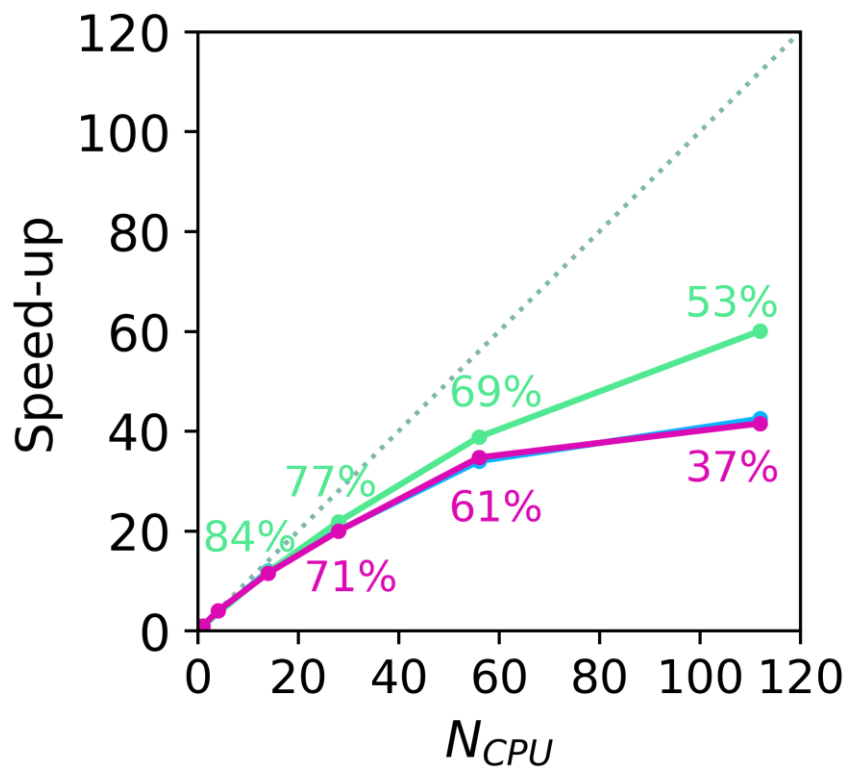
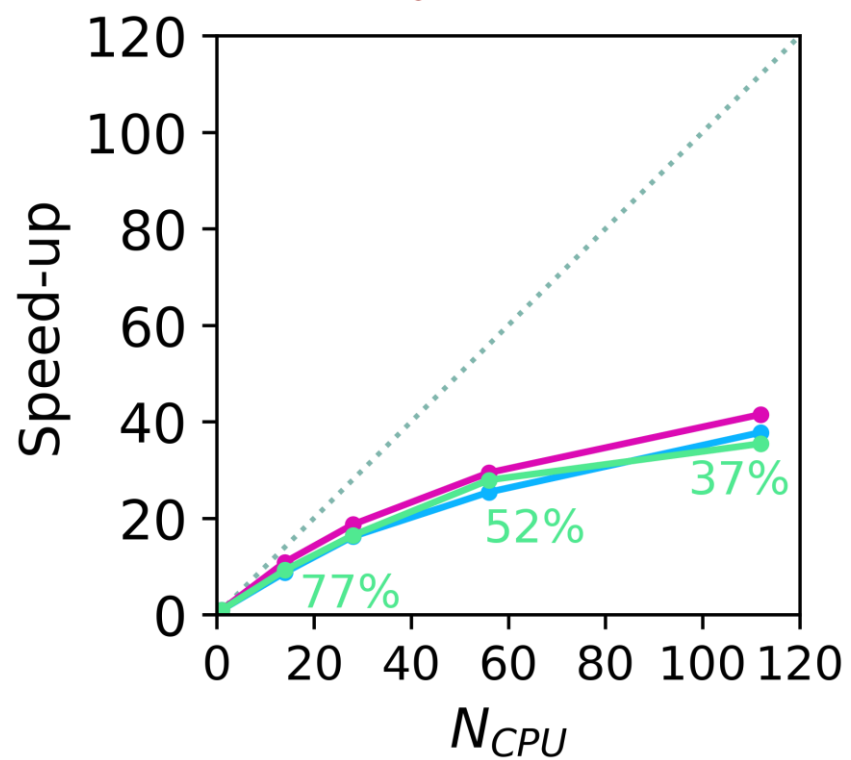
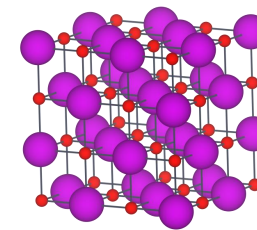
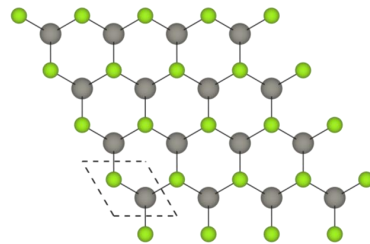
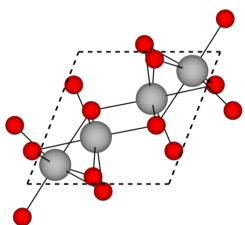


Strong Scaling

HSE06

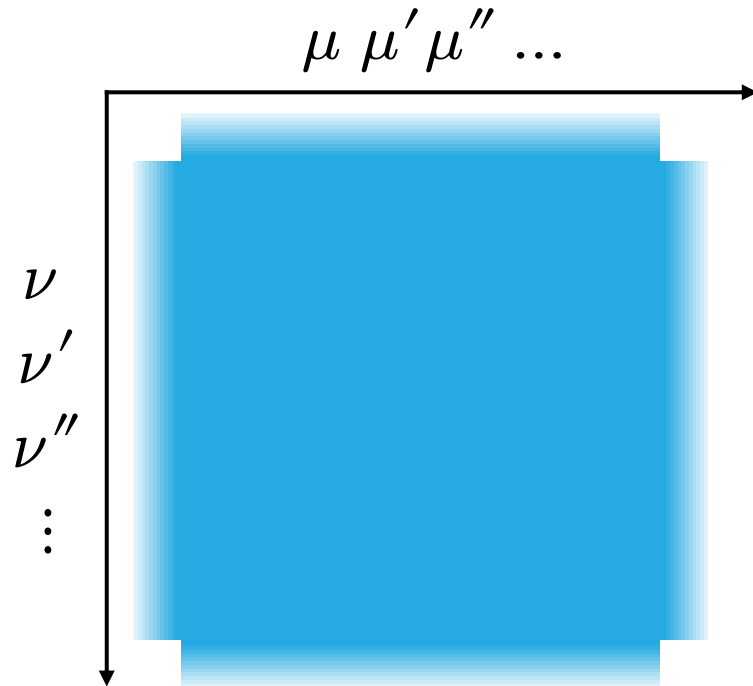
PBE - 15%

PBE



Direct Space Matrices

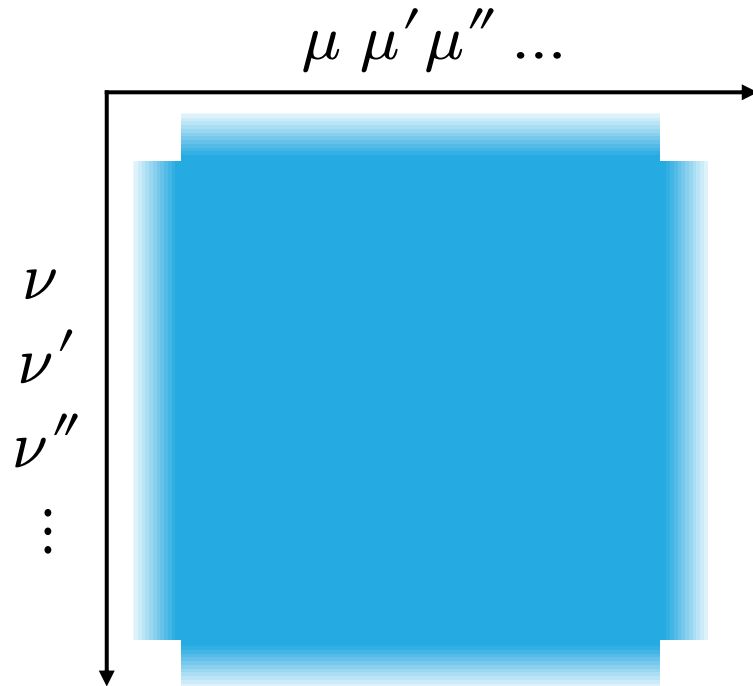
$\mu \nu$ **Atomic Orbitals** in the Reference Cell



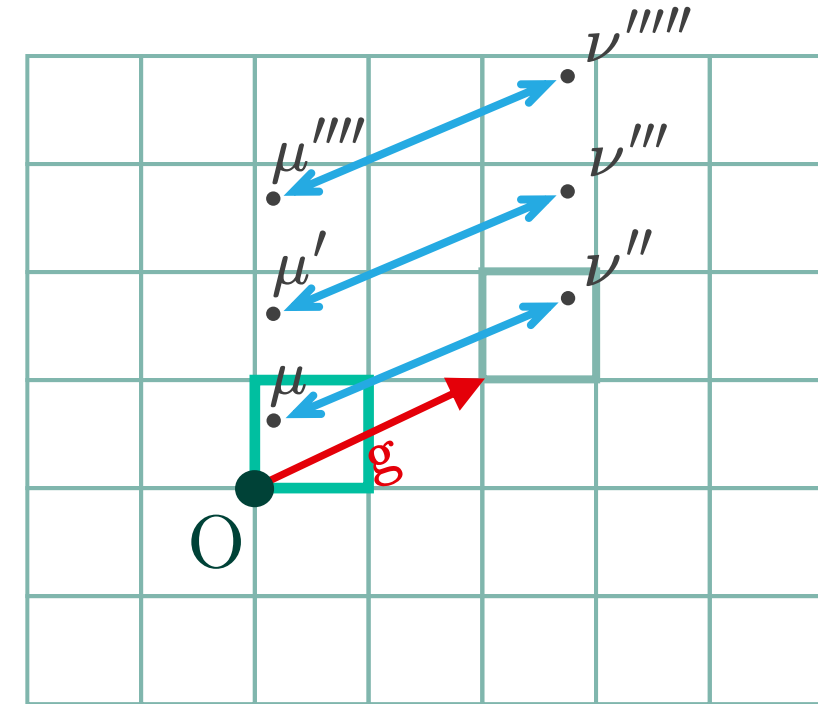
Infinite Matrix

Direct Space Matrices

$\mu \nu$ **Atomic Orbitals** in the Reference Cell



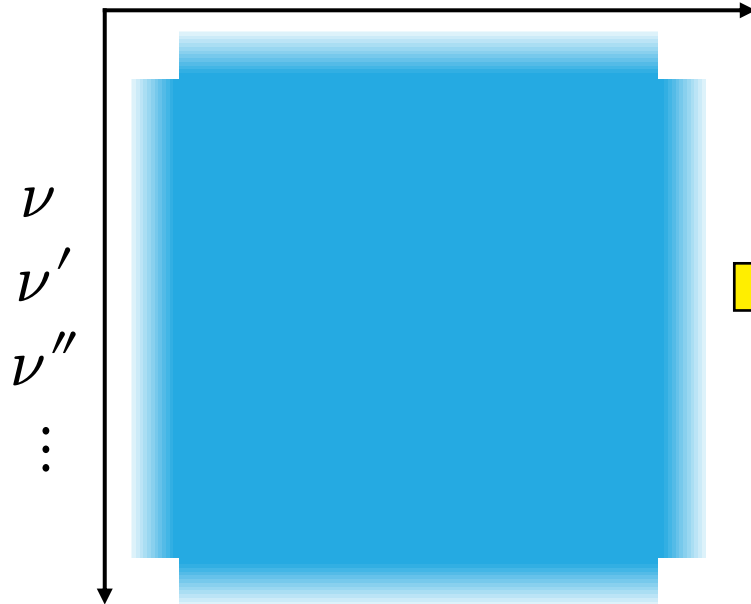
Infinite Matrix



Direct Space Matrices

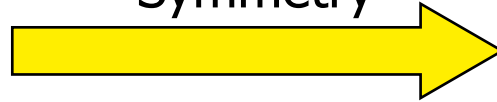
$\mu \nu$ **Atomic Orbitals** in the Reference Cell

$\mu \mu' \mu'' \dots$

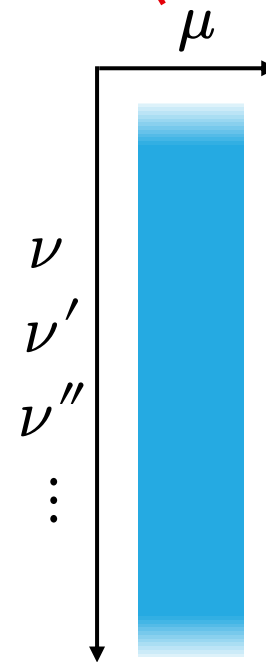


Infinite Matrix

Translational
Symmetry



All Cells



Semi-Infinite Matrix

Only in
Reference Cell

